

1.Metallic Gaskets – Product Range

S. No.	Product	Types / Variants
1	Ring Type Joint (RTJ) Gaskets	R – Oval, R – Octagonal, RX, BX, SBX, SRX
2	Rubber-Coated RTJ Gaskets	—
3	PTFE-Insert RTJ Gaskets	—
4	Protective-Coated RTJ Gaskets	—
5	Bonnet Rings	—
6	Lens Rings	—
7	Blind Gaskets	—
8	VX–AX Gaskets	—
9	IX Gaskets	—
10	Delta Gaskets	—
11	Bridgeman Gaskets	—
12	Spectacle Blind Gaskets	—
13	Transition RTJ Gaskets	—
14	Soft-Material-Serrated RTJ Gaskets	—
15	Weld-Lip Seals	—

2.Semi-Metallic Gaskets – Product Range

S. No.	Product	Types / Variants
1	Spiral Wound Gaskets	Style R, Style RIR, Style CG, Style CGI, Style CGI with Bolt Holes Full-Face
2	Kammprofile Gaskets	Type Z, Type ZA, Type ZI
3	Metal Jacketed Gaskets	Corrugated Metal Jacketed, Flat Metal Jacketed, Double Jacketed, Heat Exchanger
4	Flange Isolation Kits	<u>TX-ISOPRO™</u> , TX-CERAMPRO™ . TX Type D, <u>TX TYPE -N</u>

3.Non-Metallic Gaskets – Product Range

S. No.	Product	Types / Variants
1	Rubber Gaskets	Neoprene, Nitrile (NBR), EPDM, Silicone, Viton® (FKM), Natural Rubber, SBR, Butyl
2	PTFE Gaskets	Virgin PTFE, Expanded PTFE, Modified PTFE, Filled PTFE, PTFE Envelope, PTFE Ring, PTFE Full-Face, PTFE Envelope with Insert
3	ePTFE Gaskets	Expanded PTFE – Sheet, Joint Sealant, Cut Gaskets, Custom Profiles
4	Asbestos free Gaskets	TXCO AF-110 , TXCO AF-120 , TXCO AF-139 , TXCO AF-154 , TXCO AF-155 GF TXCO AF-159 , TXCO AF-160 , TXCO AF-170 , TXCO AF-190
5	Compressed Non Asbestos	TXCO Style 11 , TXCO Style 20, TXCO Style 39 , TXCO Style 51, TXCO Style 54 TXCO Style 55 , TXCO Style 59 Oil, TXCO Style 60 Acid

4.Steel-Inserted Gaskets

S. No.	Product	Types / Materials
1	Steel-Inserted Gaskets	Rubber, Graphite

5.Plug gasket

6. Flanges – Product Range

S. No.	Flange Type	Description / Variants
1	Weld Neck Flanges	—
2	Slip-On Flanges	—
3	Blind Flanges	—
4	Socket Weld Flanges	—
5	Lap Joint Flanges	—
6	Threaded (Screwed) Flanges	—
7	Orifice Flanges	—
8	Custom Flanges	Manufactured as per customer drawings, specifications and application requirements

7.Monolithic Insulation Joints

8. Valve components

9.commerical products

Metallic gaskets.

TXCO Sealing Products Private Limited manufactures and supplies metallic gaskets for critical applications in the oil & gas, petrochemical, refining, chemical and power industries.

Our metallic gaskets are designed for demanding services requiring high pressure and temperature resistance, mechanical strength and chemical compatibility.

We manufacture using stainless steel, duplex, super duplex, nickel alloys and other special alloys, selected to suit the application and customer requirements.

Quality is maintained through controlled material selection, dimensional accuracy, hardness, surface finish and traceability. Each gasket is manufactured and inspected in accordance with the applicable standard, drawing and customer specification.

Quality Controlled. Standards Compliant. Reliable Sealing.

<u>MATERIAL</u>	<u>UNS No.</u>	<u>Max. Brinell Hardness</u>	<u>Max. Rockwell Hardness</u>	<u>Description</u>
Soft Iron		90	56	D
Low Carbon Steel		120	68	S
4-6% Chrome 1/2% Moly	K42544	130	75	F5
Type 304 Stainless Steel	S30400	160	83	S304
Type 316 Stainless Steel	S31600	160	83	S316
Type 347 Stainless Steel	S34700	160	83	S347
Type 410 Stainless Steel	S41000	170	86	S410
Titanium Grade 2	R50400	215	94	
Alloy 600	N06600	200	92	
Alloy 625	N06625	200	92	
Alloy 800	N08800	200	92	

<u>MATERIAL</u>	<u>UNS No.</u>	<u>Max. Brinell Hardness</u>	<u>Max. Rockwell Hardness</u>	<u>Description</u>
Alloy 825	N08825	200	92	
Hastelloy	N10001	200		
Alloy C276	N10276	200	92	
SMO 254	S32154	180	89	
Zeron 100		200		
Super Duplex	S31803	200	92	
321 Stainless Steel		160	83	
MONEL 400		130	72	
309 Stainless Steel		160	83	
SMO 254		180	89	
Alloy 20		160	83	

Other exotic materials available on request.

Ring type joint

Ring Type Joint (RTJ) Gaskets are metallic sealing solutions designed for high-pressure, high-temperature and demanding flange connections. They provide a reliable metal-to-metal seal and are available in R, RX and BX profiles, selected according to flange design, pressure class and service conditions.

At TXCO Sealing Products Private Limited, RTJ gaskets are a key part of our sealing portfolio for critical oil & gas, petrochemical, refining, chemical and power applications. Our range covers various materials,

sizes and pressure ratings, with products aligned to ASME B16.20 and API 6A requirements for dependable sealing performance.

Type R – Oval

- Oval-shaped cross-section designed for line-contact sealing.
- Suitable for compatible oval and octagonal RTJ grooves.
- Commonly used in legacy equipment and low-to-medium pressure applications.
- Provides reliable sealing for conventional flange connections.
- Suitable for a wide range of industrial services where standard RTJ sealing is required.

Type R – Octagonal

- Features an octagonal cross-section for concentrated point-contact sealing.
- Designed specifically for octagonal RTJ grooves.
- Provides improved sealing performance compared with the oval profile.
- Widely used in oil & gas, petrochemical, refining and industrial applications.
- Suitable for demanding pressure and temperature conditions.

Type RX

- Features a modified octagonal cross-section with pressure-energized sealing.
- Internal pressure increases the sealing force against the groove walls.
- Designed for compatible API 6A and ASME flange connections.
- Commonly used in high-pressure valves, wellheads and pressure vessels.
- Suitable for critical services involving pressure and temperature fluctuations.
- Must be matched with the correct groove dimensions and is not interchangeable with BX.

Type BX

- High-pressure, pressure-energized RTJ gasket designed for extreme service conditions.
- Incorporates a pressure-equalizing hole to support uniform pressure distribution.
- Designed exclusively for compatible API 6A BX flanges.
- Suitable for pressures up to 20,000 psi, depending on material and application.
- Commonly used in wellheads, Christmas trees, subsea systems and high-pressure manifolds.
- Requires proper alignment, groove condition and surface finish for reliable sealing.

PTFE-Insert Ring Type Joint Gaskets

- R and RX Ring Type Joint Gaskets can be supplied with PTFE inserts to reduce turbulence within the flow path and minimize erosion of the gasket and flange bore surfaces.
- The PTFE insert forms a protective barrier between the process medium and the exposed internal surfaces of the gasket and flange, helping reduce erosion, wear and flow-induced damage.
- For RX-type RTJ gaskets, the PTFE insert incorporates strategically positioned radial pressure-passage holes. These allow system pressure to reach the sealing surfaces and preserve the self-energizing characteristics of the RX profile.
- The insert is located between the inner diameter of the RTJ gasket and the flange bore, occupying the annular gap between the gasket and flange bore.
- During flange assembly, the PTFE insert is securely retained between the mating flange faces, maintaining its position under operating conditions.
- The design provides a combination of metallic sealing strength and PTFE chemical resistance, making it suitable for demanding process environments.
- PTFE insert configurations can be selected based on gasket profile, flange geometry, operating pressure, temperature and process medium.

Rubber-Coated Ring Type Joints

- Rubber-Coated Ring Type Joints consist of a metallic RTJ core encapsulated in a nitrile rubber (NBR) coating, with the core typically manufactured from soft iron or low-carbon steel.
- Designed primarily for pressure testing applications, where protection of the flange sealing faces and RTJ grooves is important.
- The resilient rubber coating provides additional contact with the mating surfaces, supporting enhanced sealing during pressure testing.
- The coating helps minimize mechanical damage, surface marking and wear on flange faces during assembly and testing.
- By isolating the metallic core from direct metal-to-metal contact with the groove surfaces, the coating also helps reduce the potential for surface corrosion and damage.

- Available in standard RTJ sizes, with custom dimensions and configurations available to suit specific flange and testing requirements.

Protective Coatings

- Protective coatings are applied to metallic Ring Type Joint Gaskets to enhance surface protection and improve suitability for specific service conditions.
- As specified by applicable API requirements, RTJ gaskets manufactured from soft iron and low-carbon steel can be electroplated with zinc coating up to a maximum thickness of 13 µm.
- TXCO Sealing Products Private Limited has the capability to provide Zinc, Nickel and Silver electroplated coatings for RTJ gaskets.
- Zinc coating provides protection against surface oxidation and corrosion, while Nickel and Silver coatings can be selected for specific service, compatibility and sealing requirements.
- Coating selection and thickness can be provided in accordance with customer specifications, applicable standards and operating conditions.

VX–AX Gaskets

- VX–AX Gaskets are specialized sealing solutions developed for demanding oil & gas, offshore and subsea applications, combining reliable pressure sealing with application-specific performance requirements.
- VX gaskets feature a face-to-face, pressure-energized design that supports reliable sealing while allowing direct hub-to-hub contact.
- The pressure-energized geometry enhances sealing performance as system pressure increases, helping maintain joint integrity under demanding operating conditions.
- VX gaskets are suitable for wellheads, riser connectors and subsea equipment, including services involving aggressive chemicals, elevated temperatures and various process fluids.
- AX gaskets incorporate sealing and insulating elements to provide both effective sealing and electrical isolation between mating flange components.
- The electrical insulation helps minimize the risk of galvanic corrosion between dissimilar metallic components and supports the integrity of the flange connection.
- VX and AX configurations can be selected according to flange design, pressure and temperature conditions, material compatibility and service environment.

IX Ring Type Joint Gaskets

- IX Ring Type Joint Gaskets, available from IX 15 to IX 1200, are designed and supplied in accordance with the requirements of Norsok L-005.
- Gasket materials are selected in accordance with applicable Norsok M-650 and Norsok M-630 requirements for demanding offshore and subsea service.
- PTFE or Xylan coatings can be applied to provide suitable surface protection and support reliable sealing performance.
- Color coding is used to identify the gasket material type in accordance with the applicable specification.
- IX gaskets can be supplied in various material grades, sizes and coating configurations based on operating conditions and project requirements.
- Silver-coated IX gaskets can also be provided where specified by the customer or project requirements.
- Suitable for critical applications where material traceability, corrosion resistance and reliable sealing performance are essential.

Transition Ring Type Joints

- Transition Ring Type Joints are specialized sealing elements designed to connect mating RTJ flanges with different groove geometries while maintaining the same pitch circle diameter.
- They provide an effective sealing solution where the flange groove profiles do not directly match, avoiding the need for flange modification or replacement.
- These combination rings are engineered to accommodate different RTJ configurations while maintaining the required sealing integrity and dimensional compatibility.
- Available with oval and octagonal sealing profiles, depending on the mating flange configurations.
- Transition RTJs are typically application-specific components and are not directly governed by standard API or ASME RTJ specifications.
- Dimensions, materials, hardness and profile geometry are established according to the specific flange configuration and service requirements.
- Suitable for specialized applications where reliable sealing, dimensional compatibility and operational integrity are critical.

Soft Material Serrated RTJ Gaskets

- Soft Material Serrated RTJ Gaskets combine a serrated metallic core with a soft facing material to provide enhanced sealing in demanding flange connections.

- The serrated profile increases the effective contact area and helps the facing material conform to minor surface irregularities in the flange groove.
- Soft facing materials such as graphite or PTFE can be selected according to the process medium, temperature and chemical conditions.
- The metallic core provides mechanical strength and dimensional stability, while the soft facing improves sealing capability and surface conformity.
- Suitable for applications where improved sealing is required against minor flange imperfections or surface irregularities.
- Material, serration geometry and facing selection can be customized based on operating pressure, temperature, flange configuration and service conditions.
- Common applications include oil & gas, petrochemical, refining, chemical and process industries.

Bonnet Gaskets

- Bonnet Gaskets are sealing components installed between the valve body and bonnet, providing a reliable pressure boundary and preventing leakage of fluids or gases.
- They are designed to maintain sealing integrity under pressure, temperature and process-fluid conditions encountered in industrial valve applications.
- Bonnet gaskets are commonly used in gate valves and other pressure-containing valve assemblies, particularly in oil & gas, petrochemical, refining and power industries.
- Depending on the valve design and service conditions, bonnet gaskets may be supplied in RTJ and other metallic sealing configurations.
- Material selection is based on operating temperature, pressure, corrosion environment and process-fluid compatibility.
- Common materials include carbon steel, stainless steel 304/316 and other alloy grades, subject to application requirements.
- Proper control of gasket dimensions, hardness, surface finish and bonnet-groove compatibility is essential for reliable sealing performance.
- Bonnet gaskets support equipment integrity, leak prevention and safe operation in critical valve services.

Lens Gaskets

- Lens Gaskets are metallic sealing elements designed for flanged connections with spherical sealing geometries.
- Unlike conventional flat gaskets, the lens profile has a convex sealing surface that mates with the corresponding concave flange seat, creating a concentrated metal-to-metal seal.
- The spherical geometry allows controlled deformation under bolt load, producing a narrow, high-integrity sealing contact.
- This design provides reliable sealing under high pressure, elevated temperatures and severe thermal cycling.
- Lens Gaskets are primarily specified in accordance with DIN 2696, with dimensions and configurations adaptable to other flange standards where required.
- The gasket material is selected with appropriate hardness relative to the flange material, allowing deformation to occur preferentially in the gasket and minimizing damage to the flange sealing surface.
- Available materials include carbon steel, stainless steel and selected alloy grades, depending on pressure, temperature and process-media requirements.
- Common applications include chemical plants, reactors, high-pressure piping systems and other critical process equipment where sealing reliability and resistance to thermal cycling are essential.

Blind Flange Metallic Gaskets

- Blind Flange Metallic Gaskets are designed for sealing blind flange connections in high-pressure and high-temperature systems.
- They provide a secure metal-to-metal sealing interface, maintaining joint integrity under demanding operating conditions.
- Suitable for critical pressure-containing applications where reliable sealing is essential.
- Manufactured from high-strength metallic materials selected according to pressure, temperature and process-fluid requirements.
- Designed to maintain dimensional compatibility with the mating flange, supporting proper fit-up and sealing performance.
- Commonly used in oil & gas, petrochemical, refining, chemical and power applications.

Spectacle Blinds

- Spectacle Blinds are figure-eight-shaped metallic devices used for positive isolation of pipeline sections during maintenance, inspection and operational activities.

- They consist of two integral plates: an open bore for normal flow and a solid blind for complete flow isolation.
- The two plates are connected by a web, allowing the blind to be rotated between the open and closed positions without removing it from the pipeline.
- In the closed position, the solid plate provides a positive mechanical barrier, offering a higher level of isolation than conventional valves alone.
- Spectacle blinds are installed between mating flanges and require suitable gaskets on the sealing faces to achieve a leak-tight flange connection.
- Gasket selection depends on the flange facing, pressure class, temperature and process medium and may include spiral wound, RTJ or other suitable gasket configurations.
- Serrated sealing surfaces may be provided to improve gasket retention and load distribution under bolting.
- Commonly used in oil & gas, petrochemical, refining, chemical and power industries where reliable pipeline isolation is essential.

Delta Gaskets

- Delta Gaskets are pressure-energized metallic seals designed for compact, high-integrity flange connections, particularly in subsea applications.
- Their geometry utilizes system pressure to enhance the sealing force, supporting reliable metal-to-metal sealing under demanding conditions.
- Designed for deepwater, high-pressure and high-temperature environments, where conventional sealing solutions may be subject to severe operating conditions.
- Commonly used in API 17D subsea equipment, including wellhead and subsea production systems.
- The pressure-energized design helps maintain sealing integrity during pressure fluctuations and demanding service conditions.
- Manufactured from suitable metallic alloys selected according to pressure, temperature, corrosion environment and material compatibility.
- Delta Gaskets are suited to critical applications where leak prevention, mechanical integrity and long-term sealing reliability are essential.

Bridgeman Gaskets

- Bridgeman Gaskets are specialized metallic seals designed for ultra-high-pressure and high-vacuum applications where exceptional sealing integrity is required.
- The sealing mechanism relies on controlled plastic deformation of the metallic gasket under high compressive loading.
- This deformation allows the gasket to conform closely to the mating surfaces, creating a high-integrity metal-to-metal seal.
- Suitable for applications involving extreme pressure differentials, vacuum conditions and demanding thermal environments.
- Material selection is based on operating pressure, temperature, corrosion resistance and compatibility with the process medium.
- Commonly associated with research equipment, nuclear systems, aerospace applications and other specialized high-integrity systems.
- Proper control of surface finish, gasket geometry, material hardness and applied load is essential to achieve reliable sealing performance.

TXCO Weld Ring Gaskets

Dual-Ring Metallic Sealing for Critical & Semi-Permanent Flange Connections

TXCO Weld Ring Gaskets—also known as lip seal weld rings—utilize a matched pair of precision metallic rings to establish a fused, 100% leak-proof joint. During installation, each individual ring is first welded directly to its respective flange face. Once the flange connection is aligned and bolted, the outer edges of the two mating rings are seal-welded together along their outer periphery to complete the hermetic seal.

Operational Features & Advantages

- **Dual-Weld Hermetic Seal:** Combines inner structural flange welds with an outer circumferential seal weld, creating a continuous metal barrier that completely eliminates leakage pathways.
- **Severe-Service Alternative:** Engineered for extreme pressure, thermal cycling, and hazardous media applications where standard elastomeric or soft sheet gaskets risk blowout or degradation.
- **Decoupled Stress Distribution:** Flange bolts handle the mechanical line loads and structural tension, allowing the weld rings to focus purely on zero-leakage fluid containment.
- **Semi-Detachable Maintenance Profile:** Ideal for piping systems requiring infrequent maintenance. When servicing is necessary, the joint is opened by removing the flange bolts and mechanically cutting the outer peripheral weld seam without damaging the underlying flange face.

Technical Summary

Feature	Engineering Standard
Design Type	Symmetrical Dual-Ring Metallic Assembly
Joint Category	Semi-Permanent / Semi-Detachable
Primary Joining Method	Outer Edge Seal Welding (GTAW / TIG)
Recommended Service	High-Temperature Steam, Lethal Media, Infrequent Overhaul Lines

SEMI METALLIC GASKETS

Semi-metallic gaskets combine metallic strength with compressible sealing materials to provide reliable performance under demanding pressure, temperature and process conditions.

TXCO Sealing Products Private Limited offers Spiral Wound, Kammprofile and Metal Jacketed Gaskets, along with Flange Isolation Kits for sealing and electrical isolation. Material and construction are selected according to flange design, service conditions and process media.

Designed for critical applications across oil & gas, petrochemical, refining, chemical and power industries, TXCO solutions deliver dependable sealing, mechanical stability and service reliability.

Spiral Wound Gaskets

Spiral Wound Gaskets are semi-metallic sealing elements designed for reliable performance under demanding pressure, temperature and process conditions. Their alternating metallic winding and filler construction provides flexibility, recovery and resistance to operating fluctuations.

At TXCO Sealing Products Private Limited, Spiral Wound Gaskets are designed and configured in accordance with applicable ASME B16.20, ASME B16.5 and ASME B16.47 requirements, with construction and materials selected based on flange configuration, process medium, pressure, temperature and service conditions.

Construction & Materials		
Component	Typical Materials	Function
Winding	SS 304, SS 316, SS 321, SS 347, Inconel, Monel, Nickel Alloys	Provides strength, resilience and recovery

Filler	Flexible Graphite, PTFE, Ceramic, Mica	Provides the primary sealing medium
Inner Ring	Carbon Steel, SS 304/316, SS 321/347, Nickel Alloys	Prevents inward buckling and protects the winding
Outer Ring	Carbon Steel, SS 304/316, SS 321/347	Provides centering and compression control
Special Alloys	Duplex, Super Duplex, Hastelloy, Inconel	For corrosive and severe service conditions

Key Construction Elements

- **Outer Ring:** Provides accurate centering, alignment and controlled compression during flange assembly.
- **Inner Ring:** Supports the winding, prevents inward buckling and helps protect against process turbulence and erosion.
- **Sealing Element:** Alternating metal winding and filler layers form the primary sealing barrier.
- **Material Selection:** Winding, filler and ring materials are selected according to pressure, temperature, chemical compatibility and flange requirements.

Style R – Basic Spiral Wound Gasket

- Consists of the spiral wound sealing element without inner or outer guide rings.
- Manufactured using alternating layers of metallic winding strip and filler material.
- Additional winding at the start and end provides stability and resistance to unwinding.
- The V-shaped winding profile provides spring-like recovery during pressure and temperature variations.
- Suitable for tongue & groove, male & female and other confined flange configurations.
- Useful where the flange design itself provides gasket positioning and compression control.

Style RIR – Inner Ring

- Consists of a spiral wound sealing element with an integral inner metal ring.
- The inner ring provides radial support and additional structural stability.
- Helps prevent inward buckling or collapse of the winding under compression.
- Acts as a barrier against process-media turbulence, erosion and debris accumulation.

- Provides additional control of the gasket's inner sealing diameter.
- Commonly used with male & female, spigot-and-recess and confined flange configurations.

Style CG – Outer Ring

- Consists of the spiral wound sealing element with an outer centering and guide ring.
- The outer ring ensures accurate gasket positioning relative to the flange bore.
- Functions as a compression limiter, helping prevent excessive crushing of the winding.
- Supports uniform gasket loading during bolt tightening and flange assembly.
- Commonly used with raised face (RF) and flat face (FF) flanges.
- Suitable for general process piping and moderate service conditions.

Style CGI – Inner & Outer Rings

- Incorporates the spiral wound sealing element with both inner and outer metallic rings.
- The inner ring provides radial reinforcement, compression control and media protection.
- Helps minimize inward buckling, turbulence and erosion at the gasket ID.
- The outer ring provides centering, alignment and controlled compression.
- Provides improved mechanical stability under high pressure and temperature conditions.
- Recommended for severe service, aggressive media and critical flange connections.
- Suitable for applications requiring enhanced sealing reliability and protection of the winding element.

Style CGI – Bolt-Hole Type

- Incorporates the CGI construction with bolt holes provided in the outer centering ring.
- Bolt holes provide positive gasket location and alignment during flange assembly.
- Helps prevent gasket displacement or rotation during installation.
- Supports consistent positioning of the sealing element relative to the flange and bolt circle.
- The inner ring provides protection against inward buckling, erosion and process-media effects.
- The outer ring provides compression control and accurate centering.
- Suitable for critical applications where precise gasket positioning and repeatable assembly are required.

Kammprofile Gaskets

Kammprofile Gaskets are semi-metallic sealing elements consisting of a serrated metallic core with a facing material. The controlled serration profile concentrates the sealing load while the facing material fills minor surface irregularities, providing reliable sealing under demanding pressure, temperature and process conditions.

At TXCO Sealing Products Private Limited, Kammprofile Gaskets are configured according to flange design, operating conditions and process media, with materials selected for the required service. Designs can be supplied to applicable ASME B16.20 and project-specific requirements.

Type Z – Basic Kammprofile

- Consists of a serrated metallic sealing core without an integral guide ring.
- The serrations provide controlled contact and support the sealing material under compression.
- Can be supplied with suitable facing materials such as flexible graphite or PTFE, depending on service requirements.
- Suitable for applications where controlled compression and reliable metal-core support are required.
- Material selection is based on pressure, temperature and chemical compatibility.

Type ZA – Kammprofile with Outer Ring

- Incorporates a serrated sealing core with an outer centering ring.
- The outer ring provides accurate gasket positioning and alignment during flange assembly.
- Helps control compression and protects the sealing element during installation.
- Suitable for raised-face and other compatible flange configurations.
- Commonly selected for general industrial and demanding process applications.

Type ZI – Kammprofile with Inner & Outer Rings

- Features a serrated metallic core with both inner and outer rings.
 - The inner ring provides additional radial support and protects the sealing element from process turbulence and erosion.
 - The outer ring provides centering and compression control.
 - Offers enhanced mechanical stability for high-pressure, high-temperature and aggressive service conditions.
 - Suitable for critical applications where sealing integrity, stability and long service life are essential.
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Metal Jacketed Gaskets

Metal Jacketed Gaskets are semi-metallic sealing elements consisting of a soft filler enclosed within a metallic jacket. The metal jacket provides mechanical protection and resistance to pressure and temperature, while the filler accommodates flange irregularities and supports sealing performance.

At TXCO Sealing Products Private Limited, Metal Jacketed Gaskets are configured in different constructions and materials to suit heat exchangers, pressure vessels, piping and other critical process equipment. Material selection is based on the process medium, operating temperature, pressure and flange configuration.

Types of Metal Jacketed Gaskets

1. Corrugated Metal Jacketed Gaskets

- Incorporate a corrugated metallic jacket around the filler material.
- Corrugations provide increased flexibility and improved surface conformity.
- Suitable for heat exchangers, boilers and high-temperature applications.
- Available with suitable metallic jackets and fillers according to service conditions.

2. Flat Metal Jacketed Gaskets

- Consist of a flat metallic jacket enclosing a soft filler.
- Provide a robust sealing solution for conventional flange and equipment connections.
- Suitable for moderate pressure and temperature applications.
- Commonly used in piping, covers, heat exchangers and process equipment.

3. Double Jacketed Gaskets

- Manufactured with two metallic jackets enclosing the filler material.
- The double-jacket construction provides enhanced mechanical protection and sealing stability.
- Suitable for demanding pressure and temperature conditions.
- Commonly used for heat exchangers, pressure vessels and equipment flanges.

4. Heat Exchanger Gaskets

- Designed specifically for heat exchanger covers, channels and tube-sheet assemblies.
 - Available in configurations suited to complex flange geometries and multiple sealing requirements.
 - Can incorporate different jacket materials, fillers, pass partitions and sealing arrangements.
 - Designed to withstand thermal cycling, pressure fluctuations and process-fluid exposure.
 - Material selection is based on the shell-side and tube-side operating conditions
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Flange Isolation Kits

TXCO Flange Isolation Kits are designed to electrically isolate flanged joints and help prevent galvanic corrosion between dissimilar metals. They also support cathodic protection systems by preventing unwanted electrical continuity across the flange connection.

A typical kit includes an insulating gasket, insulating sleeves, insulating washers and steel backup washers. The insulating components work together to provide effective electrical separation while maintaining the required sealing integrity and mechanical stability of the flange joint.

These kits are widely used in oil & gas, petrochemical, refining, chemical, water and process industries, where corrosion control and reliable flange isolation are essential.

TX-ISOPRO™

Engineered for critical galvanic corrosion control and Cathodic Protection (CP) integrity, the TX ISOPRO™ series delivers total dielectric isolation and zero-leakage fluid containment across oil & gas, petrochemical, and process piping.

Combining a high-strength Glass Reinforced Epoxy (GRE G10/G11) substrate with an integrated TX-FLEX™ modified ePTFE primary seal, the system eliminates creep relaxation and resists aggressive media (pH 0–14) under thermal cycling. Paired with insulating sleeves and double washers, it creates a complete 360° electrical barrier.

Available Configurations

- TX ISOPRO™-F (Type F – Raised Face): Fits inside the bolt circle across the raised face. Simplifies centering and minimizes exposure to external contaminants in RF flange connections.
- TX ISOPRO™-E (Type E – Full Face): Extends to the flange outer diameter with precision bolt holes. Delivers full face-to-face coverage to prevent metallic debris from bridging the isolation gap.

TX-CERAMPRO™

TX-CERAMPRO™ is a fire-safe flange isolation gasket designed to provide electrical insulation and reliable sealing in critical flange connections. TX-CERAMPRO™ redefines high-temperature pipeline integrity through a revolutionary dual-sheet facing technology, establishing a new standard where absolute thermal endurance meets uncompromised dielectric defense. Engineered around a precision steel core encapsulated within high-strength Glass Reinforced Epoxy (GRE), it integrates proprietary **TX-CERAMSHIELD™** and **TX-IGNISIL™** facings to create an impenetrable barrier built for the world's most punishing process lines.

Engineering Highlights

- **Revolutionary Dual-Sheet Facing:** Fuses specialized ceramic and mineral matrices to deliver continuous thermal management up to 1000°C with zero carbon oxidation or material degradation.
- **API 6FB Certified Fire Safety:** Formulated to preserve joint tightness and structural seal integrity under direct flame exposure and catastrophic thermal shock.
- **Permanent Dielectric Defense:** High-density GRE encapsulation isolates stray electrical currents, halting galvanic corrosion across mating flanges for the lifetime of the connection.
- **Low-Emission Dual Sealing:** Synergized mica and PTFE sealing profiles deliver micro-fine fluid containment, excelling in sour gas (H_2S), high-pH media, and volatile hydrocarbons.
- **Complete System Integration:** Fully compatible with Raised Face (RF), Full Face (FF), and Ring Type Joint (RTJ) flanges, supplied with matching high-dielectric sleeves and isolation washers.

TX Type D – RTJ Isolation Gasket

- Designed specifically for Ring Type Joint flange connections where electrical isolation is required.
- Installed within the RTJ groove to provide both metallic sealing and electrical separation between the mating flanges.
- The insulating sealing element is selected to withstand the required pressure, temperature and process-media conditions.
- Particularly suitable for high-integrity systems where cathodic protection and galvanic corrosion prevention are important.
- Requires proper compatibility between the gasket, RTJ groove geometry, flange materials and isolation components for reliable performance.

TX TYPE -N :Flange Isolation Gasket

TXCO Type N is a flange isolation system designed to provide electrical isolation and reliable sealing in RF and FF flange connections. The gasket uses modified, bi-directionally oriented PTFE as the primary sealing and insulating element, offering strong chemical resistance and improved resistance to PTFE relaxation and cold flow.

- The PTFE sealing element provides electrical insulation and fluid sealing within the flange joint.
- Suitable for a broad range of chemical services, with compatibility across approximately pH 0–14, subject to specific media compatibility.
- Designed for reliable performance under pressure, temperature and chemical exposure.
- Suitable for RF and FF flange connections.
- Applicable for service temperatures up to 200°C, with pressure ratings selected according to the applicable flange class.

- GRE G10 (FR4) insulating sleeves and washers are recommended for enhanced mechanical and electrical performance.
- GRE G10 offers low moisture absorption compared with conventional phenolic materials, helping maintain insulation resistance and mechanical stability throughout service.
- Suitable for oil & gas, chemical, petrochemical, water and process applications where electrical isolation and chemical resistance are required.

Non metallic Gaskets

Non-metallic gaskets conform to flange surfaces and minor irregularities, creating a reliable barrier against leakage. Their flexibility, chemical resistance and material compatibility make them suitable for varied process conditions.

At TXCO Sealing Products Private Limited, materials such as rubber, PTFE and compressed fibre are selected according to the fluid, pressure, temperature and service environment—because a good seal must remain dependable when the system is under pressure.

Rubber Gaskets

Rubber Gaskets provide flexible, resilient sealing by conforming to flange surfaces and accommodating minor irregularities. Their ability to maintain contact under compression makes them suitable for sealing liquids and gases across a wide range of industrial applications.

At TXCO Sealing Products Private Limited, rubber compounds are selected according to the medium, temperature, pressure and service environment, with different elastomers available for specific requirements.

Types of Rubber Gaskets

- **Neoprene:** Good resistance to weathering, ozone and oxidation; suitable for outdoor and general industrial applications.
- **NBR (Nitrile):** Excellent resistance to oils, fuels and hydraulic fluids; widely used in oil & gas and industrial service.
- **EPDM:** Strong resistance to water, steam, weathering, UV and many chemicals.
- **Silicone:** Excellent flexibility and temperature stability; suitable for applications requiring cleanliness and electrical insulation.
- **FKM (Viton®):** High resistance to heat, fuels, oils and aggressive chemicals for demanding process conditions.
- **Natural Rubber:** High elasticity, resilience and abrasion resistance; suited to selected water and general service applications.

- **SBR:** Good abrasion resistance and economical performance, particularly for water and general industrial service.
- **Butyl:** Very low gas permeability with excellent resistance to moisture and air, making it suitable for gas-sealing applications.

Expanded PTFE Gaskets

Expanded PTFE (ePTFE) Gaskets are soft, chemically resistant sealing elements designed to conform closely to flange surfaces and compensate for surface irregularities and flange imperfections. Their low permeability, chemical inertness and low seating stress make them suitable for demanding sealing applications.

At TXCO Sealing Products Private Limited, ePTFE gaskets are selected for applications requiring chemical resistance, reliable sealing and low bolt loads, particularly where conventional gasket materials may be unsuitable.

Key Features

- Excellent resistance to acids, alkalis, solvents and aggressive chemicals.
- Wide operating temperature capability with good sealing stability.
- Highly conformable, helping seal irregular and damaged flange surfaces.
- Requires relatively low seating stress, reducing flange loading.
- Non-stick surface reduces adhesion and facilitates gasket removal.
- Good resistance to creep and long-term compression effects.
- Suitable for chemical, pharmaceutical, food, power and process industries.
- Available as sheets, tapes, cut gaskets and application-specific shapes.

Steel Insert - Gaskets

Steel Insert -Rubber Gaskets

Rubber Gaskets combine a lens-shaped elastomeric sealing body with an internal steel reinforcement ring, creating a robust composite sealing element for demanding flange applications.

- The rubber covering is firmly vulcanised to the steel insert, forming an integrated structure with high mechanical stability.
- The steel insert improves blow-out resistance and dimensional stability under pressure.
- The lens-shaped profile concentrates contact pressure at the sealing area, enhancing sealing performance.

- The elastomer provides flexibility and surface conformity, helping accommodate minor flange irregularities.
- The composite construction is suitable for applications requiring reliable sealing, mechanical strength and resistance to pressure loading.
- Rubber compound and steel grade can be selected according to the process medium, pressure, temperature and service environment.

Steel Insert Graphite Gaskets

Steel Insert Graphite Gaskets combine expanded flexible graphite with a metallic reinforcement core, providing a robust sealing element for demanding flange applications. The steel insert improves mechanical strength, rigidity and resistance to blowout, while the graphite conforms to mating surfaces to create an effective seal.

- ☐ Expanded graphite conforms to mating surfaces and provides excellent sealing, thermal stability and chemical resistance.
- ☐ The steel insert improves rigidity, dimensional stability and resistance to blowout during pressure and temperature cycling.
- ☐ Suitable for high-pressure, high-temperature and thermal cycling applications.
- ☐ Offers excellent resistance to steam, hot water, gases and many process fluids.
- ☐ Free from resins, impregnations, organic binders and fibre fillers.
- ☐ Suitable for oil & gas, petrochemical, refining, chemical and power industries, where dependable sealing and long service life are required.

Plug Gaskets

Plug Gaskets are sealing elements designed to provide a reliable seal between a plug and its mating surface, preventing leakage of fluids or gases in industrial equipment and piping systems.

- Provide secure sealing under varying pressure and temperature conditions.
- Available in suitable metallic and non-metallic materials based on the application.
- Material selection can be matched to process media, chemical exposure and operating conditions.
- Designed for simple installation and replacement, helping reduce maintenance downtime.
- Suitable for applications requiring reliable containment and resistance to pressure fluctuations, temperature changes and corrosive environments.

- Commonly used across oil & gas, petrochemical, chemical, refining and general process industries.

ASBESTOS FREE GASKETS:

TXCO AF-110

TXCO AF-110 is a general-purpose non-asbestos compressed fiber gasket engineered for reliable fluid and gas containment in applications with low internal pressure and low-to-moderate operating temperatures. Formulated from high-grade synthetic organic fibers bound with elastomeric binders, it is specifically optimized for sealing flanges under low bolt-loading conditions. This versatile jointing media provides a cost-effective, asbestos-free solution for plant maintenance, utility lines, and OEM equipment without compromising joint integrity.

Performance Highlights

- **Dual-Matrix Structural Integrity:** Engineered in non-metallic and steel-wire mesh reinforced variants to eliminate creep relaxation and resist blowout under dynamic pressure surges.
- **Anti-Friction Graphited Finish:** Factory-applied release coating prevents metal-to-gasket adhesion, eliminating flange face pitting and accelerating overhaul turnarounds.
- **Low-Seating Stress Sealability:** High compressibility profile achieves gas-tight sealing under low bolt torque—ideal for cast iron, light-duty, or distorted flange faces.
- **Broad Media Containment:** Superior chemical resistance against hot/cold water, low-pressure steam, fuel oils, synthetic lubricants, and aliphatic hydrocarbons.

Technical Specifications

Specification Parameter	Non-Metallic Variant	Metallic Variant (Wire-Reinforced)
Maximum Operating Pressure	40 Bar (4.0 MPa / ~580 PSI)	50 Bar (5.0 MPa / ~725 PSI)
Max Continuous Service Temp.	220°C	220°C
Max Short-Term Peak Temp.	280°C	300°C
Finish / Identity Colour	Brown	Brown
Design Standards	BS 7531:2006 Grade-Y / IS 17796:2022 Grade-Y	BS 7531:2006 Grade-Y / IS 17796:2022 Grade-Y

Target Applications

- **Utility & Process Piping:** Potable/cooling water loops, low-pressure steam headers, and HVAC fluid circuits.
- **Industrial Machinery & OEM:** Gearbox covers, pump casings, inspection hatches, and light-duty valve bodies.
- **Hydrocarbon Infrastructure:** Fuel oil transfer networks, lube oil circuits, and industrial gas distribution manifolds.

TXCO AF-120

TXCO AF-120 is an advanced non-asbestos compressed fiber gasket manufactured from high-tensile aramid fibers blended with functional inorganic fillers and an elastomeric nitrile rubber (NBR) binder matrix. Engineered to maintain continuous seating stress and resist torque loss under thermal cycling, this flexible gasket delivers dependable static sealing across fluid, gas, steam, and hydrocarbon flange connections. TXCO AF-120 is supplied in both standard General Purpose (GP) cut gaskets and steel wire-mesh reinforced Metallic gasket configurations.

Key Technical Highlights

- **Resilient Aramid-NBR Matrix:** Engineered with durable aramid fibers and synthetic elastomers to form a robust gasket structure that resists creep relaxation and bolt torque loss.
- **Wire-Reinforced Steam Gasket:** The Metallic grade incorporates an embedded high-tensile steel wire mesh insert, boosting pressure containment to 60 Bar to prevent blowout in dynamic steam and vapour lines.
- **Anti-Stick Surface Coating:** Coated with a smooth, factory-applied graphited release finish that prevents gasket adhesion to flange faces, enabling fast overhaul turnarounds and protecting seating surfaces from pitting.
- **Elevated Thermal Endurance:** Capable of continuous operation up to 250°C (482°F) with peak thermal surge capabilities reaching 400°C (725°F).

Technical Specifications

Operating Parameter	TXCO AF-120 GP Gasket	TXCO AF-120 Metallic Gasket
Maximum Operating Pressure	50 Bar (5.0 MPa / 725 PSI)	60 Bar (6.0 MPa / 870 PSI)
Max Continuous Service Temp.	250°C (482°F)	250°C (482°F)
Peak Short-Term Temperature	400°C (725°F)	400°C (725°F)
Gasket Architecture	Aramid Fibers + NBR Matrix	Steel Wire Gauze + Aramid/NBR Matrix
Surface Finish & Identity	Green or Graphited Black (Anti-Stick)	Graphited Black (Anti-Stick)
Specification Compliance	BS 7531:2006 Grade-Y / IS 17796:2022 Grade-Y	BS 7531:2006 Grade-Y / IS 17796:2022 Grade-Y

Primary Target Applications

- **Steam & Vapour Lines:** Saturated steam manifolds, condensate recovery piping, and process heating circuits (Metallic gasket recommended).
- **Process Utilities:** High-pressure water distribution, compressed air systems, inert gas headers, and cooling circuits.
- **Machinery & Valve Flanges:** Centrifugal pump casings, valve bonnet joints, heat exchangers, boiler inspection covers, and pipe manifolds.

TXCO AF-139

TXCO AF-139 is a premium, universal non-asbestos compressed fiber gasket engineered for high-integrity static sealing across severe thermal and high-pressure duties. Manufactured from high-tensile compressed aramid fibers blended with advanced functional inorganic fibers and bound within a resilient nitrile rubber (NBR) matrix, it delivers exceptional torque retention and structural stability in aggressive steam, oil, and chemical environments. TXCO AF-139 is supplied in both standard Non-Metallic cut gaskets and steel wire-mesh reinforced Metallic gasket configurations.

Key Technical Highlights

- **Grade-X Structural Resilience:** Fully compliant with BS 7531 Grade-X standards, delivering maximum resistance to creep relaxation, matrix crushing, and bolt torque loss under dynamic operational loads.
- **Wire-Reinforced High-Pressure Gasket:** The Metallic grade incorporates an embedded steel wire gauze insert, elevating pressure containment up to 100 Bar to prevent gasket blowout in extreme steam lines and pulsation-heavy piping.
- **Anti-Stick Surface Coating:** Coated with a smooth, factory-applied graphited release finish (available in Orange or Graphited Black) that eliminates flange face adhesion, preventing seating surface corrosion and speeding up overhaul turnarounds.
- **Universal Chemical & Thermal Defense:** Superior sealing performance against continuous high-temperature steam, synthetic lubricants, fuels, hydrocarbons, and mild chemical process fluids.

Technical Specifications

Specification Parameter	TXCO AF-139 Non-Metallic Gasket	TXCO AF-139 Metallic Gasket
Maximum Operating Pressure	90 Bar (9.0 MPa / ~1305 PSI)	100 Bar (10.0 MPa / ~1450 PSI)
Max Continuous Service Temp.	250°C (450°F)	250°C (450°F)
Peak Short-Term Temperature	400°C (720°F)	400°C (720°F)
Gasket Architecture	Aramid Fibers + NBR Matrix	Steel Wire Gauze + Aramid/NBR Matrix
Surface Finish & Identity	Orange or Graphited Black (Anti-Stick)	Orange or Graphited Black (Anti-Stick)
Specification Compliance	BS 7531:2006 Grade-X / IS 17796:2022 Grade-X	BS 7531:2006 Grade-X / IS 17796:2022 Grade-X

Primary Target Applications

- **High-Pressure Steam & Vapour Loops:** Superheated steam headers, boiler feed manifolds, and high-duty condensate return lines (Metallic gasket recommended).
- **Petrochemical & Oil Processing:** Hydrocarbon transport lines, fuel handling networks, hot oil circuits, and synthetic lubricant manifolds.

- **Heavy Equipment & Power Utilities:** Boiler doors, turbine casings, shell-and-tube heat exchangers, and high-pressure process valve body joints.

TXCO AF-154

TXCO AF-154 is an ultra-high-performance, non-asbestos compressed fiber gasket engineered for extreme mechanical and thermal stresses in critical flanged joints. Manufactured from superior heat-resistant aramid fibers blended with functional inorganic fibers and high-density nitrile elastomeric (NBR) binders, it easily surpasses BS 7531 Grade-X requirements. Available in standard High-Pressure Non-Metallic (Yellow) and steel wire-mesh reinforced Metallic (Graphited Black) configurations, TXCO AF-154 provides unmatched blowout resistance and structural integrity across severe industrial process conditions.

Key Technical Highlights

- **BS 7531 Grade-X Certification:** Delivers elite creep-relaxation resistance, superior stress retention, and zero matrix crushing under maximum bolt torque loads.
- **Wire-Reinforced Ultra-Pressure Barrier:** The Metallic variant incorporates a high-tensile steel wire insertion, boosting hydraulic containment up to 160 Bar (2321 PSI) to prevent catastrophic blowout in dynamic steam loops.
- **Extreme Thermal Endurance:** Engineered for continuous operational temperatures up to 300°C (572°F), with short-term thermal surge capabilities reaching 450°C for non-metallic and 475°C for metallic variants.
- **Anti-Friction Surface Finish:** Factory-applied graphited release finish on the metallic grade prevents metal-to-gasket fusion, preserving flange faces and accelerating plant overhaul cycles.

Technical Specifications

Specification Parameter	TXCO AF-154 High Pressure (Non-Metallic)	TXCO AF-154 Metallic (Wire-Reinforced)
Maximum Operating Pressure	150 Bar (15.0 MPa / ~2175 PSI)	160 Bar (16.0 MPa / ~2321 PSI)
Max Continuous Service Temp.	300°C (572°F)	300°C (300°C / 572°F)
Max Short-Term Peak Temp.	450°C (810°F)	475°C (887°F)
Gasket Architecture	Superior Heat-Resistant Aramid + NBR Matrix	Steel Wire Gauze + Aramid/NBR Matrix
Identity Colour & Finish	Yellow (Anti-Stick Finish)	Graphited Black with Anti-Stick Finish
Specification Compliance	BS 7531:2006 Grade-X / IS 17796:2022 Grade-X	BS 7531:2006 Grade-X / IS 17796:2022 Grade-X

Target Applications

- **Extreme Steam & Power Generation:** Superheated steam headers, high-pressure boiler feed manifolds, steam turbines, and main steam isolation flanges.
- **Refinery & Chemical Process Units:** High-pressure hydrocarbon headers, hot thermal oil circuits, aggressive solvent lines, and gas processing lines.
- **High-Load Mechanical OEM:** Pressure vessel inspection ports, high-pressure heat exchangers, heavy centrifugal pump casings, and autoclave closures.

TXCO AF-155 GF

TXCO AF-155 GF is an elite, non-asbestos compressed fiber gasket engineered for maximum thermal endurance and structural integrity in severe-service flange connections. Manufactured from an advanced multi-fiber blend—combining high-tensile aramid, glass, and mineral fibers bound within a Hydrogenated Nitrile (HNBR) elastomer matrix—it maintains critical bolt torque retention and superior sealability across aggressive chemical and steam services. Available in standard Non-Metallic and steel wire-mesh reinforced Metallic variants.

Key Technical Highlights

- **Advanced Quad-Fiber HNBR Matrix:** Fuses aramid, glass, and inorganic mineral fibers with high-grade HNBR rubber to deliver exceptional tensile strength, zero matrix crushing, and minimal creep relaxation.
- **Extreme Thermal Operating Envelope:** Delivers continuous operating endurance up to 400°C (Non-Metallic) and 450°C (Metallic), with peak short-term thermal surge capabilities reaching 500°C.
- **High-Pressure Wire Insertion:** The Metallic variant incorporates an embedded steel wire gauze insert, raising pressure containment to 160 Bar to prevent blowout under dynamic pressure waves.
- **BS 7531 Grade-X Certification:** Fully meets Grade-X standards, providing resistance to thermal degradation, blowout, and chemical attack from dilute acids, alkalies, and hydrocarbons.

Technical Specifications

Specification Parameter	TXCO AF-155 GF Non-Metallic Gasket	TXCO AF-155 GF Metallic Gasket
Maximum Operating Pressure	150 Bar (15.0 MPa / ~2175 PSI)	160 Bar (16.0 MPa / ~2320 PSI)
Max Continuous Service Temp.	400°C	450°C
Max Short-Term Peak Temp.	450°C	500°C
Max Continuous Steam Temp.	375°C	425°C
Gasket Composition	Aramid + Glass + Mineral Fibers + HNBR	Steel Wire Gauze + Multi-Fiber/HNBR

Specification Parameter	TXCO AF-155 GF Non-Metallic Gasket	TXCO AF-155 GF Metallic Gasket
Surface Finish & Identity	Purple (Anti-Stick Finish)	Purple or Graphited Black (Anti-Stick)
Specification Compliance	BS 7531:2006 Grade-X	BS 7531:2006 Grade-X

Primary Target Applications

- **Superheated Steam & Power Utilities:** High-pressure steam headers, boiler feed manifolds, power plant piping, and steam turbine flanges.
- **Petrochemical & Chemical Units:** Fertilizer plants, hot hydrocarbon transfer, organic solvents, and dilute acid or alkali chemical lines.
- **Heavy Industrial & Marine Equipment:** Shipbuilding piping manifolds, automotive exhaust/engine joints, and high-duty pressure vessels.

TXCO AF-159

TXCO AF-159 is a severe-service non-asbestos compressed aramid fiber gasket engineered for critical fluid, gas, and hydrocarbon containment under demanding mechanical and thermal loads. Manufactured from high-grade heat-resistant aramid fibers blended with functional inorganic fillers and bound in a premium nitrile rubber (NBR) matrix, TXCO AF-159 delivers exceptional torque retention, oil-swell resistance, and gas-tight performance. Offered in both standard Non-Metallic and steel wire-mesh reinforced Metallic configurations, it fully complies with BS 7531 Grade-X standards.

Performance Highlights

- **Heavy-Duty NBR Matrix:** Formulated specifically to resist fuel penetration, fluid absorption, and chemical swelling across high-load mechanical drive systems.
- **Steel-Wire Blowout Protection:** The Metallic variant features an embedded high-tensile steel wire mesh insert, boosting pressure containment to 160 Bar (2321 PSI) to prevent gasket blowout under dynamic pressure spikes.
- **BS 7531 Grade-X Compliance:** Delivers superior creep-relaxation resistance and stress retention, ensuring stable seating stress under cyclic thermal operation up to 300°C.
- **Anti-Stick Surface Finish:** Coated with a smooth graphited release layer (in Black or Graphited Black) that prevents metal-to-gasket fusion, preserving flange faces and expediting disassembly during maintenance overhauls.

Technical Specifications

Specification Parameter	TXCO AF-159 Non-Metallic Gasket	TXCO AF-159 Metallic Gasket
Maximum Operating Pressure	150 Bar (15.0 MPa / ~2175 PSI)	160 Bar (16.0 MPa / ~2321 PSI)
Max Continuous Service Temp.	300°C (540°F)	300°C (540°F)

Specification Parameter	TXCO AF-159 Non-Metallic Gasket	TXCO AF-159 Metallic Gasket
Max Short-Term Peak Temp.	450°C (810°F)	450°C (810°F)
Gasket Architecture	Heat-Resistant Aramid + NBR Matrix	Steel Wire Gauze + Aramid/NBR Matrix
Identity Colour & Finish	Black or Graphited Black (Anti-Stick)	Black or Graphited Black (Anti-Stick)
Specification Compliance	BS 7531:2006 Grade-X / IS 17796:2022 Grade-X	BS 7531:2006 Grade-X / IS 17796:2022 Grade-X

Target Applications

- **Engines & Power Generation:** Diesel engine blocks, power generator sets, fuel injection manifolds, and heavy machinery housings.
- **Hydraulic & Fluid Machinery:** High-pressure hydraulic pumps, fluid power systems, lubricant filtration units, and fuel distribution lines.
- **Industrial Process Utilities:** Fuel oil headers, lubrication circuits, chemical service manifolds, and plant steam loops.

TXCO AF-160

TXCO AF-160 Acid is a specialized chemical- and acid-resistant non-asbestos compressed fiber gasket engineered for high-integrity static sealing in aggressive fluid and gas environments. Formulated from premium high-tensile aramid fibers blended with performance-enhancing additives and a chemical-resistant elastomeric binder matrix (including high-grade NBR), it resists degradation, swelling, and embrittlement when exposed to acidic solutions and harsh process chemicals.

Key Technical Highlights

- **Acid & Chemical Defense:** Engineered with specialized binders and additives to deliver exceptional chemical resistance against organic and inorganic acids, alkalis, and reactive process fluids.
- **BS 7531 Grade-X Compliance:** Fully meets Grade-X performance requirements, delivering high stress-retention capabilities, low seating torque demands, and superior creep relaxation resistance.
- **100 Bar Pressure Containment:** Handles working pressures up to 100 Bar (1450 PSI), providing a robust safety margin against gasket blowout and fugitive emissions in high-pressure chemical piping.
- **Distinct Off-White Identification:** Clean Off-White color profile ensures instant visual verification on the job site, preventing material misapplication in critical process lines.

Technical Specifications

Specification Parameter	TXCO AF-160 Acid Gasket Performance
Maximum Operating Pressure	100 Bar (10.0 MPa / ~1450 PSI)
Max Continuous Service Temp.	250°C (482°F)
Max Short-Term Peak Temp.	300°C (572°F)
Gasket Composition	Aramid Fibers + Specialized Additives + Chemical-Grade NBR Binder
Identity Colour	Off-White
Specification Compliance	BS 7531:2006 Grade-X / IS 17796:2022 Grade-X

Primary Target Applications

- **Chemical & Petrochemical Processing:** Transfer lines for dilute/moderate acids, alkaline wash headers, solvent circuits, and aggressive chemical process piping.
- **Fertilizer & Pickling Operations:** Acid dosing pumps, pickling bath piping manifolds, reaction vessels, and chemical storage tanks.
- **Industrial Water & Effluent Care:** Acidic wastewater treatment loops, neutralization systems, and hazardous chemical recovery units.

TXCO AF-170

TXCO AF-170 is an ultra-high-performance non-asbestos compressed fiber gasket engineered for the most demanding mechanical and thermal duties across critical industrial flange connections. Manufactured from an advanced multi-fiber matrix—blending high-tensile aramid, mineral, and carbon fibers with expanded graphite bound within a premium Hydrogenated Nitrile (HNBR) elastomer matrix—it delivers unrivaled torque retention, micro-sealability, and oxidation resistance. TXCO AF-170 is supplied in standard High-Pressure Non-Metallic and steel wire-mesh reinforced Metallic gasket configurations.

Key Technical Highlights

- **Advanced Carbon-Graphite Multi-Fiber Matrix:** Fuses aramid, carbon, and mineral fibers with expanded graphite and HNBR rubber to provide extreme structural strength and virtually zero creep relaxation.
- **200 Bar Ultra-Pressure Barrier:** The Metallic variant incorporates high-tensile steel wire mesh insertion, raising hydraulic pressure containment up to 200 Bar (~2900 PSI) to prevent blowout under violent shockwaves.
- **Extreme Thermal Envelope:** Operates continuously at temperatures up to 425°C (Non-Metallic) and 525°C (Metallic), with short-term peak thermal surge capabilities reaching 650°C.
- **BS 7531 Grade-X Certification:** Delivers elite resistance against matrix crushing, bolt torque loss, thermal degradation, and chemical attack from oils, aggressive solvents, and volatile gases.

Technical Specifications

Specification Parameter	TXCO AF-170 Non-Metallic Gasket	TXCO AF-170 Metallic Gasket
Maximum Operating Pressure	160 Bar (16.0 MPa / ~2320 PSI)	200 Bar (20.0 MPa / ~2900 PSI)
Max Continuous Service Temp.	425°C	525°C
Max Short-Term Peak Temp.	600°C	650°C
Max Continuous Steam Temp.	450°C	550°C
Gasket Composition	Aramid + Mineral + Carbon Fibers + Graphite + HNBR	Steel Wire Gauze + Multi-Fiber/Graphite/HNBR
Identity Colour & Finish	Black (Anti-Stick Finish)	Black with Anti-Stick Finish
Specification Compliance	BS 7531:2006 Grade-X	BS 7531:2006 Grade-X

Primary Target Applications

- **Superheated Steam & Power Utilities:** Ultra-high-pressure steam headers, boiler feed manifolds, steam turbines, and power plant main steam lines.
- **Petrochemical & Chemical Refining:** Hot oil circuits, organic solvents, gas processing units, fertilizer plants, and aggressive hydrocarbon headers.
- **Heavy OEM & Internal Combustion:** Heavy IC engine blocks, generator housings, automotive exhaust manifolds, shipbuilding propulsion piping, and high-duty pressure vessels.

TXCO AF-190

TXCO AF-190 is a severe-service non-asbestos compressed fiber gasket designed for multi-media sealing across extreme thermal and pressure environments. Engineered from a blend of premium synthetic aramid fibers, high-modulus carbon, and heat-dissipating graphite bound within a high-performance Hydrogenated Nitrile (HNBR) elastomer matrix, TXCO AF-190 delivers exceptional chemical resistance, minimal stress relaxation, and superior torque retention. It is available in standard Non-Metallic (Performance) and stainless steel wire-reinforced Metallic gasket configurations.

Key Technical Highlights

- **Aramid-Carbon-Graphite HNBR Matrix:** Synergizes thermal-dissipating graphite, high-modulus carbon, and aramid fibers with an HNBR elastomer binder to prevent binder hardening and matrix degradation under severe thermal stress.

- **Stainless Steel Wire Blowout Guard:** The Metallic grade incorporates an embedded stainless steel wire mesh insertion, raising maximum pressure capability to 170 Bar (2465 PSI) for unmatched blowout security in high-impulse media lines.
- **BS 7531 Grade-X Compliance:** Fully meets BS 7531 Grade-X standards, providing low seating stress requirements and high creep-relaxation resistance across dynamic piping networks.
- **Anti-Stick Surface Technology:** Pre-treated with a factory-applied release coating to prevent gasket-to-flange bonding, simplifying cleanup during routine plant overhauls and preserving flange seating faces.

Technical Specifications

Specification Parameter	TXCO Performance (Non-Metallic)	AF-190 (Non-Metallic)	TXCO AF-190 Metallic (Wire-Reinforced)
Maximum Operating Pressure	150 Bar (15.0 MPa / ~2175 PSI)		170 Bar (17.0 MPa / ~2465 PSI)
Max Continuous Service Temp.	300°C (572°F)		300°C (572°F)
Max Short-Term Peak Temp.	500°C (932°F)		500°C (932°F)
Gasket Composition	Aramid + Carbon + Graphite + HNBR Matrix		Stainless Steel Wire Gauze + Aramid/Carbon/Graphite/HNBR
Identity Colour & Finish	Dark Grey / Black (Anti-Stick Finish)		Graphited Black (Anti-Stick Finish)
Specification Compliance	BS 7531:2006 Grade-X / IS 17796:2022 Grade-X		BS 7531:2006 Grade-X / IS 17796:2022 Grade-X

Target Applications

- **Thermal & Power Infrastructure:** Superheated steam headers, high-temperature heat exchangers, condensate recovery lines, and boiler equipment.
- **Hydrocarbon & Petrochemical Processing:** Fuel transport networks, hot oil loops, organic solvents, industrial gases, and synthetic refrigerants.
- **Universal Industrial Machinery:** High-duty centrifugal pumps, valve bonnet flanges, compressors, and severe-service process piping manifolds (except concentrated acids).

Compressed Non-Asbestos Fiber(CNAF) Gaskets:

TXCO Style 11

TXCO Style 11 is a versatile compressed asbestos fiber (CAF) gasket engineered for reliable static sealing in medium-duty industrial applications operating under low-to-moderate pressures and temperatures. Manufactured from processed asbestos fibers intimately bound with high-quality rubber compounds, it exhibits excellent conformability and sealing performance under light bolt-load conditions. TXCO Style 11 is available in standard Non-Metallic and steel wire-mesh reinforced Metallic gasket configurations, each finished with a smooth anti-stick graphite surface treatment.

Key Technical Highlights

- **Low Bolt-Load Sealability:** High compressibility profile allows the gasket to deform efficiently under low torque, ensuring a tight seal on light-duty, cast iron, or slightly distorted flange surfaces.
- **Steel-Wire Blowout Defense:** The Metallic variant incorporates an embedded steel wire gauze matrix, elevating internal pressure resistance to 30 Bar (435 PSI) to protect against pressure spikes.
- **Anti-Stick Surface Treatment:** Factory-applied graphite release coating prevents gasket fusion to flange faces, protecting seating surfaces and reducing cleanup time during maintenance turnarounds.
- **Dual-Grade Standard Compliance:** Fully meets IS 2712-1971 Grade C and IS 2712-1998 Grade W/3 standards for general-purpose utility jointing.

Technical Specifications

Specification Parameter	TXCO Style 11 Non-Metallic Gasket	TXCO Style 11 Metallic Gasket
Maximum Operating Pressure	25 Bar (2.5 MPa / 362 PSI)	30 Bar (3.0 MPa / 435 PSI)
Max Continuous Service Temp.	270°C (486°F)	300°C (572°F)
Gasket Composition	Chrysotile Asbestos Fibers + Elastomeric Binder	Steel Wire Mesh + Asbestos/Elastomer Matrix
Surface Coating & Finish	Smooth Anti-Stick Graphite Coating	Smooth Anti-Stick Graphite Coating
Specification Compliance	IS 2712-1971 Grade C / IS 2712-1998 Grade W/3	IS 2712-1971 Grade C / IS 2712-1998 Grade W/3

Primary Target Applications

- **Utility & Low-Pressure Piping:** Cold/hot water loops, low-pressure air lines, drainage manifolds, and inert gas service.
- **Light Industrial Equipment:** Cast iron valve flanges, pump casings, gearbox inspection hatches, and low-pressure fluid tanks.
- **Plant Maintenance:** Cost-effective general-purpose sealing across utility networks requiring low bolt torque

TXCO Style 20

TXCO Style 20 is a medium-duty compressed asbestos fiber (CAF) gasket engineered for static sealing across general-purpose liquid, gas, and vapour systems. Formulated from processed chrysotile asbestos fibers bonded within a quality elastomeric binder matrix, it deforms effectively under low-to-moderate bolt torque to seal micro-imperfections on flange surfaces. Available in standard Non-Metallic Red/Graphited Black and steel wire-mesh reinforced Metallic configurations, TXCO Style 20 delivers balanced structural integrity and thermal endurance for plant utility maintenance.

Key Technical Highlights

- **Wire-Reinforced Steam Capability:** The Metallic grade incorporates an embedded steel wire gauze insert that raises internal pressure containment to 40 Bar (580 PSI) and continuous thermal tolerance to 415°C, making it ideal for general-purpose steam and vapour manifolds.
- **Low Bolt-Load Sealing:** High compressibility profile achieves gas-tight seating stress under low bolt torque—ideal for cast iron, light-duty, or slightly worn flange faces.
- **Anti-Stick Surface Coating:** Factory-applied graphited release finish prevents gasket-to-metal fusion, preserving seating faces and speeding up cleanup during plant overhaul cycles.
- **IS 2712 Compliance:** Fully conforms to IS 2712-1971 Grade C and IS 2712-1998 Grade W/3 manufacturing specifications for medium-duty jointing.

Technical Specifications

Specification Parameter	TXCO Style 20 Red Gasket (Non-Metallic)	TXCO Style 20 Metallic Gasket
Maximum Operating Pressure	35 Bar (3.5 MPa / ~507 PSI)	40 Bar (4.0 MPa / ~580 PSI)
Max Continuous Service Temp.	380°C (716°F)	415°C (779°F)
Gasket Core Architecture	Chrysotile Asbestos Fibers + Elastomeric Binder	Steel Wire Gauze + Asbestos/Elastomer Matrix
Identity Colour & Finish	Red or Graphited Black (Anti-Stick Finish)	Graphited Black (Anti-Stick Finish)
Special Sheet Dimensions	Up to 2000 mm x 4500 mm	Standard & Custom Cut Sizes
Specification Compliance	IS 2712-1971 Grade C / IS 2712-1998 Grade W/3	IS 2712-1971 Grade C / IS 2712-1998 Grade W/3

Primary Target Applications

- **Steam & Vapour Lines:** Medium-pressure steam headers, condensate return piping, and process heating vapour circuits (Metallic gasket recommended).
- **General Industrial Utilities:** Plant water distribution, low-pressure gas mains, compressed air lines, and drainage manifolds.
- **Equipment Flanges:** Cast iron valve bodies, centrifugal pump casings, gearbox covers, and low-pressure fluid storage tanks.

TXCO Style 39

TXCO Style 39 is a high-duty compressed asbestos fiber (CAF) gasket engineered to maintain static sealability under intense pressure and elevated thermal loads across refinery, petrochemical, and distillate processing applications. Formulated from select chrysotile asbestos fibers compounded with oil-resistant elastomeric binders, it delivers superior stress retention, high tensile strength, and resistance to hydrocarbon degradation. TXCO Style 39 is available in standard Non-Metallic and steel wire-mesh reinforced Metallic gasket configurations.

Key Technical Highlights

- **Hydrocarbon & Solvent Resistance:** Formulated with high-grade elastomeric binders that prevent fluid penetration, structural breakdown, and gasket swelling in distillate and refinery lines.
- **Steel-Wire Blowout Security:** The Metallic variant incorporates an embedded steel wire mesh gauze, raising pressure containment to 100 Bar (1450 PSI) to guard against dynamic pressure spikes.
- **Elevated Thermal Capability:** Engineered for continuous operational temperatures reaching 440°C in non-metallic and 480°C in metallic variants without significant bolt torque loss.
- **Anti-Stick Graphite Finish:** Coated with a smooth, factory-applied graphited release finish to prevent metal-to-gasket fusion, preserving flange faces and shortening overhaul downtime.

Technical Specifications

Specification Parameter	TXCO Style 39 Non-Metallic Gasket	TXCO Style 39 Metallic Gasket
Maximum Operating Pressure	80 Bar (8.0 MPa / 1160 PSI)	100 Bar (10.0 MPa / 1450 PSI)
Max Continuous Service Temp.	440°C (824°F)	480°C (896°F)
Gasket Core Architecture	Chrysotile Fibers + Oil-Resistant Matrix	Steel Wire Gauze + Asbestos/Elastomer Matrix
Identity Colour & Finish	Orange/Grey or Graphited Black (Anti-Stick)	Graphited Black (Anti-Stick Coating)
Specification Compliance	IS 2712-1998 Grade O/2	IS 2712-1998 Grade O/2

Primary Target Applications

- **Petrochemical & Refinery Systems:** Catalytic cracking units, fuel oil manifolds, distillate transport lines, and heavy hydrocarbon headers.
- **High-Thermal Process Utilities:** Hot thermal fluid loops, superheated transfer circuits, and high-pressure steam distribution lines.
- **Heavy Process Equipment:** Shell-and-tube heat exchangers, valve bonnet joints, boiler feed pumps, and reaction vessel flanges.

TXCO Style 51

TXCO Style 51 is a high-pressure compressed asbestos fiber (CAF) gasket engineered to deliver robust static sealing under elevated temperatures and high operational pressures. Manufactured from selected heat-resistant chrysotile asbestos fibers bound within a high-grade elastomeric compound matrix, it provides exceptional torque retention and structural resistance against flange degradation in refinery, petrochemical, and process line duties. TXCO Style 51 is offered in standard Non-Metallic and steel wire-mesh reinforced Metallic gasket configurations.

Key Technical Highlights

- **High-Pressure Structural Core:** Formulated to withstand severe mechanical seating loads while maintaining a tight fluid and gas barrier under system pressure fluctuations.
- **Steel-Wire Blowout Protection:** The Metallic variant incorporates an embedded high-tensile steel wire mesh insert, elevating internal pressure capability to 100 Bar (1450 PSI) to guard against dynamic pressure surges.
- **Elevated Thermal Endurance:** Capable of continuous operational performance up to 440°C (Non-Metallic) and 480°C (Metallic) without experiencing matrix thermal breakdown or bolt torque loss.
- **Anti-Stick Surface Coating:** Supplied with a smooth factory-applied graphited release finish that prevents metal-to-gasket fusion, protecting seating faces and speeding up cleanup during overhaul turnarounds.

Technical Specifications

Specification Parameter	TXCO Style 51 Non-Metallic Gasket	TXCO Style 51 Metallic Gasket
Maximum Operating Pressure	80 Bar (8.0 MPa / 1160 PSI)	100 Bar (10.0 MPa / 1450 PSI)
Max Continuous Service Temp.	440°C (824°F)	480°C (896°F)
Gasket Core Architecture	Chrysotile Asbestos Fibers + Elastomeric Binder	Steel Wire Gauze + Asbestos/Elastomer Matrix
Identity Colour & Finish	Brown/Grey or Graphited Black (Anti-Stick)	Graphited Black (Anti-Stick Coating)

Specification Parameter	TXCO Style 51 Non-Metallic Gasket	TXCO Style 51 Metallic Gasket
Specification Compliance	IS 2712-1998 Grade W/2	IS 2712-1998 Grade W/2

Primary Target Applications

- **Refinery & Petrochemical Processing:** Distillate transfer piping, hydrocarbon process headers, crude oil circuits, and solvent manifolds.
- **High-Pressure Steam Utilities:** Superheated steam mains, boiler feed lines, condensate recovery headers, and high-duty plant steam connections.
- **Heavy Process Equipment:** Shell-and-tube heat exchangers, high-pressure pump casings, valve bonnet joints, and pressure vessel inspection ports.

TXCO Style 54 Super

TXCO Style 54 Super is an elite compressed asbestos fiber (CAF) gasket engineered for critical static sealing in extreme high-temperature and high-pressure steam environments. Formulated to handle superheated steam lines, power boiler headers, and heavy petrochemical services, it resists structural crushing and maintains tight bolt stress under harsh thermal loads. Available in standard Non-Metallic and wire-reinforced Metallic configurations (incorporating a minimum 18 mesh x 30 gauge steel wire insert), TXCO Style 54 Super sets the benchmark for severe-duty utility sealing.

Key Technical Highlights

- **Steel Wire Mesh Insertion:** The Metallic grade incorporates a heavy-duty steel wire gauze core (minimum 18 mesh x 30 gauge), raising pressure resistance to 160 Bar (2330 PSI) to prevent gasket blowout in high-impulse steam networks.
- **Extreme Thermal Operating Limit:** Delivers continuous operational endurance up to 550°C (1022°F) in non-metallic variants and 600°C (1112°F) in wire-reinforced metallic variants.
- **Anti-Stick Surface Treatment:** Factory-applied graphited release finish prevents gasket adhesion to flange faces at elevated temperatures, preserving seating surfaces and expediting cleanup during maintenance turnarounds.
- **Severe Media Containment:** Maintains gas-tight sealing against superheated steam, cryogenic air separation gases, petroleum distillates, and combustion gases.

Technical Specifications

Specification Parameter	TXCO Style 54 Super Non-Metallic Gasket	TXCO Style 54 Super Metallic Gasket
Maximum Operating Pressure	150 Bar (15.0 MPa / 2175 PSI)	160 Bar (16.0 MPa / 2330 PSI)
Max Continuous Service Temp.	550°C (1022°F)	600°C (1112°F)

Specification Parameter	TXCO Style 54 Super Non-Metallic Gasket	TXCO Style 54 Super Metallic Gasket
Reinforcement Gauze Spec.	Standard Asbestos Matrix	Min. 18 Mesh x 30 Gauge Steel Wire
Identity Colour & Finish	Yellow/Grey or Graphited Black (Anti-Stick)	Graphited Black with Anti-Stick Coating

Primary Target Applications

- **Boilers & Superheated Steam:** Steam drums, superheater headers, main steam isolation valves, and high-pressure steam distribution mains.
- **Refinery & Petrochemical Processing:** Petroleum distillate headers, oil refinery processing units, and high-temperature hydrocarbon manifolds.
- **Heavy Propulsion & Industrial OEM:** Marine propulsion engines, heavy diesel generator sets, and air separation equipment.

TXCO Style 55

TXCO Style 55 is an advanced compressed fiber gasket engineered for extreme mechanical environments subject to intense pressure swings, rapid thermal cycling, and continuous structural vibration. Designed with exceptional dimensional stability to maintain seating stress even on narrow flange faces, it delivers leak-free containment across high-stress utility lines, engines, and heavy power equipment in both standard Non-Metallic and stainless-steel wire-reinforced Metallic variants.

Key Technical Highlights

- **Dynamic Load Stability:** Absorbs mechanical shocks, pressure surges, and continuous vibration without matrix cracking or bolt seating relaxation.
- **Narrow Flange Optimization:** High-density fiber construction ensures superior torque retention and dimensional integrity on reduced flange contact widths.
- **Stainless Steel Gauze Core (Metallic Variant):** Reinforced with a stainless steel wire mesh insertion (minimum 18 mesh x 30 gauge), elevating working pressure tolerance to 225 Bar (3260 PSI) to eliminate blowout risks.
- **Anti-Friction Graphited Coating:** Pre-treated with a factory-applied anti-stick graphited release layer that prevents metal-to-gasket fusion and protects flange seating surfaces during overhauls.

Technical Specifications

Operating Parameter	TXCO Style 55 Universal (Non-Metallic)	TXCO Style 55 Metallic (Reinforced)
Maximum Operating Pressure	200 Bar (20.0 MPa / 2900 PSI)	225 Bar (22.5 MPa / 3260 PSI)
Max Continuous Service Temp.	600°C (1112°F)	600°C (1112°F)

Operating Parameter	TXCO Style 55 Universal (Non-Metallic)	TXCO Style 55 Metallic (Reinforced)
Internal Reinforcement	Standard High-Density Matrix	Min. 18 Mesh x 30 Gauge Stainless Steel Wire
Identity Colour & Finish	Dark Grey	Graphited Black with Anti-Stick Finish
Media Suitability	Steam, Vapour, Industrial Gases, Hydrocarbons	High-Temp Steam, Vapour, Oil Loops (Excl. Acids)

Primary Target Applications

- **Engine & Compressor Systems:** Engine cylinder head joints, heavy-duty diesel generators, and high-vibration gas compressor headers.
- **Steam & Power Equipment:** High-pressure steam boilers, superheated steam manifolds, turbine housing flanges, and main steam lines.
- **High-Impulse Machinery:** Industrial oil distribution circuits, vibration-prone process piping, and narrow-profile valve flanges.

TXCO Style 59 Oil

TXCO Style 59 Oil is a high-performance compressed fiber jointing sheet engineered specifically for demanding, elevated-temperature, and high-pressure fluid applications. Formulated to withstand aggressive industrial fluids—including fuel oils, synthetic lubricants, greases, refined petrochemicals, and organic solvents—it maintains high torque retention and dimensional stability under severe mechanical loads. TXCO Style 59 Oil Premium is available in standard Non-Metallic and steel wire-mesh reinforced Metallic gasket configurations.

Key Technical Highlights

- **Hydrocarbon & Solvent Resistance:** Formulated with high-density oil-resistant binders that guard against structural degradation, binder leaching, and volumetric swelling in solvent and petroleum loops.
- **Steel-Wire Blowout Security:** The Metallic variant features an embedded high-tensile steel wire mesh insertion, boosting pressure containment capability to 160 Bar (2320 PSI) to mitigate risk during pressure surges.
- **High Thermal Endurance:** Continuous service rating reaches up to 550°C (1022°F) in non-metallic variants and 600°C (1112°F) in wire-reinforced metallic variants without thermal matrix breakdown.
- **Anti-Stick Graphite Finish:** Pre-treated with a smooth factory-applied graphited release coating to prevent gasket-to-flange adhesion, protecting flange faces and speeding up overhaul schedules.

Technical Specifications

Specification Parameter	TXCO Style 59 Oil Premium (Non-Metallic)	TXCO Style 59 Oil Premium Metallic
Maximum Operating Pressure	150 Bar (15.0 MPa / 2175 PSI)	160 Bar (16.0 MPa / 2320 PSI)
Max Continuous Service Temp.	550°C (1022°F)	600°C (1112°F)
Internal Reinforcement	Standard High-Density Matrix	Steel Wire Gauze Insertion
Identity Colour & Finish	Dark Grey	Dark Grey / Graphited Black (Anti-Stick)
Media Compatibility	Oils, Greases, Lubricants, Solvents, Petrochemicals	Heavy Hydrocarbons, High-Temp Solvents, Lube Lines

Primary Target Applications

- **Petrochemical & Oil Refining:** Solvent extraction plants, high-temperature fuel oil headers, lube oil loops, and refined hydrocarbon piping.
- **Industrial Chemical Processing:** Organic solvent transport, thermal fluid heating loops, grease delivery manifolds, and distillate process lines.
- **Heavy Industrial Machinery:** High-pressure hydraulic systems, gearbox housings, turbine lubrication circuits, and industrial oil pumps.

TXCO Style 60 Acid

TXCO Style 60 Acid is a specialized compressed jointing sheet engineered specifically to handle aggressive acid streams and highly corrosive chemical media in chemical processing industries. Formulated from acid-resistant chrysotile fibers combined with chemically inert elastomeric binders, it maintains dependable bolt loading and structural integrity without suffering binder attack or micro-porosity leaks over extended service life.

Key Technical Highlights

- **Aggressive Acid Resistance:** Specially compounded matrix resists chemical breakdown when exposed to mineral acids, organic acids, and acidic process reagents.
- **High-Pressure Reliability:** Sustains continuous operational pressures up to 150 Bar (2175 PSI) without suffering radial deformation or edge extrusion under flange torque.
- **IS 2712 Grade A/1 Compliance:** Manufactured in strict compliance with IS 2712-1971 Grade A/1 standards for acid-resistant compressed asbestos jointing sheets.
- **Visual Line Verification:** Supplied in a distinct Cream/Light Grey color scheme to enable rapid visual identification by maintenance crews, preventing accidental misapplication in chemical piping networks.

Technical Specifications

Specification Parameter	TXCO Style 60 Acid Gasket
Maximum Operating Pressure	150 Bar (15.0 MPa / 2175 PSI)
Max Continuous Service Temp.	210°C (410°F)
Gasket Core Architecture	Acid-Resistant Fibers + Inert Elastomeric Binder
Identity Colour	Cream / Light Grey
Specification Compliance	IS 2712-1971 Grade A/1
Media Suitability	Organic Acids, Mineral Acids, Corrosive Chemicals, Acidic Gases

Primary Target Applications

- **Chemical Processing Plants (CPI):** Acid transfer lines, chemical reactor flanges, acid recycling loops, and neutralization manifolds.
- **Fertilizer & Petrochemical Facilities:** Sulfuric acid, phosphoric acid, and nitric acid dosing systems and acid byproduct transport piping.
- **Industrial Processing Equipment:** Acid storage tank hatches, chemical metering pumps, pickling line piping, and glass-lined equipment flanges.

TXCO INDUSTRIAL FLANGES

Precision-engineered flange solutions for demanding industrial applications

ABOUT TXCO

TXCO Sealing Products Private Limited is a manufacturer of industrial sealing products and precision-engineered components serving the oil & gas, petrochemical, refining, chemical, power and process industries.

TXCO manufactures and supplies industrial flanges in stainless steel, duplex, super duplex and nickel-based and special alloys, in accordance with applicable international standards, approved drawings and customer or project specifications.

TXCO operates under certified quality management systems and is committed to manufacturing in accordance with applicable ASME and API requirements. Our quality systems, manufacturing controls and inspection processes are aligned with the requirements of the relevant product and project specifications.

Our flange manufacturing focuses on dimensional accuracy, material integrity, controlled machining, traceability and dependable sealing performance. Our experience in industrial gaskets also gives us a practical understanding of the flange, gasket and bolting interface.

PRECISION ENGINEERED. STANDARDS DRIVEN. QUALITY CONTROLLED.

WHAT IS A FLANGE?

A flange is a mechanical component used to create a bolted, detachable connection between piping and equipment such as pipes, valves, pumps, pressure vessels, heat exchangers and other process equipment.

A typical flanged joint consists of a flange, gasket, bolting and mating flange. The flanges provide the mechanical connection and transmit the required bolt load, while the gasket is compressed between the mating faces to create a pressure-tight seal.

Reliable flange performance depends on material, dimensions, pressure class, facing, surface finish, bore and bolt-hole accuracy, RTJ groove geometry where applicable, gasket compatibility and operating conditions.

FLANGE TYPES

1. WELD NECK FLANGES

High-integrity flanges for critical service

Weld Neck Flanges have a long, tapered hub and are connected to the pipe by butt welding. The tapered hub provides a gradual transition between the flange and pipe and helps distribute stresses at the flange-to-pipe connection. The bore can be matched to the pipe bore to maintain a smooth flow path and reduce turbulence and localized erosion.

Typical applications:

- Oil & gas
- Refineries and petrochemical plants
- Chemical processing
- Power generation
- Pressure vessels and heat exchangers
- High-pressure and high-temperature piping

2. SLIP-ON FLANGES

Practical flange connections for industrial piping

Slip-On Flanges fit over the outside diameter of the pipe and are normally welded internally and externally. They offer a practical and economical connection where the service conditions permit their use. They are commonly selected for general industrial and utility piping.

Typical applications:

- Process piping
- Utility systems
- General industrial piping
- Equipment connections
- Water and utility services

3. BLIND FLANGES

Positive closure and mechanical isolation

A Blind Flange is a solid flange with bolt holes but no central bore. It is used to close the end of a piping system, valve, pressure vessel or equipment nozzle. Blind flanges provide positive isolation and allow a connection to be reopened for inspection, maintenance, testing or future expansion.

Typical applications:

- Pipeline isolation

- Pressure vessel nozzles
- Equipment connections
- Hydrostatic testing
- Maintenance and shutdowns
- Pipeline termination

4. SOCKET WELD FLANGES

Compact flanges for small-bore piping

Socket Weld Flanges have a machined socket into which the pipe is inserted before welding. They are primarily used for smaller-diameter piping and are suitable for selected high-pressure applications within the applicable design rating.

Typical applications:

- Small-bore piping
- High-pressure piping
- Instrumentation systems
- Chemical processing
- Hydraulic systems
- Process piping

5. LAP JOINT FLANGES

Flexible alignment for maintenance-intensive systems

Lap Joint Flanges are used with a compatible Stub End. The pipe is welded to the stub end while the flange remains free to rotate around it, allowing convenient bolt-hole alignment during assembly. This configuration is useful where frequent dismantling or maintenance is required.

Typical applications:

- Chemical processing
- Corrosive services
- Stainless-steel piping
- Duplex and special-alloy piping
- Process plants
- Maintenance-intensive systems

6. THREADED (SCREWED) FLANGES

Flanged connections without welding

Threaded Flanges, also known as Screwed Flanges, have internal threads that engage with the external threads of the pipe. They can be installed without flange-to-pipe welding and are used for selected applications where welding is impractical, undesirable or restricted. They are generally considered for selected low-pressure or utility services, subject to design requirements.

Typical applications:

- Utility systems
- Instrumentation
- Small-diameter piping
- Selected process applications
- Applications where welding is restricted

7. ORIFICE FLANGES

Precision flanges for flow measurement

Orifice Flanges are specialized flange assemblies used in differential-pressure flow measurement. They incorporate provisions for pressure tapping on either side of an orifice plate. The pressure differential across the orifice is used to determine the process-fluid flow rate.

Typical applications:

- Oil & gas
- Refineries
- Petrochemical plants
- Chemical processing
- Process plants
- Flow measurement systems

FLANGE FACING TYPES

The flange face forms the interface between the flange and gasket. Correct face configuration, dimensions and surface condition are important to the performance of the bolted joint.

Face Type	Description	Typical Application
RF	Raised Face	General industrial and process applications
FF	Flat Face	Selected equipment and piping applications
RTJ	Ring Type Joint	High-pressure and high-temperature applications

Raised Face (RF)

Raised Face flanges have a raised gasket seating surface above the main flange face and are widely used in industrial and process piping.

Flat Face (FF)

Flat Face flanges have the gasket seating surface in the same plane as the flange face and are used where a flat-face connection is specified.

Ring Type Joint (RTJ)

RTJ Flanges have a precision-machined annular groove for a metallic Ring Type Joint gasket. They are commonly used in demanding high-pressure and high-temperature services. TXCO's RTJ gasket experience provides a practical understanding of groove dimensions, gasket geometry, hardness and surface condition.

ISO CERTIFIED. ASME & API STANDARDS DRIVEN.

TXCO is an ISO-certified organization with a strong quality-management approach covering manufacturing, inspection, traceability and continual improvement. Our flange products are manufactured and inspected in accordance with the applicable ASME and API standards, customer drawings and project specifications.

For each requirement, the applicable standard, material specification, pressure class, dimensions, tolerances, facing and inspection requirements are reviewed before production.

STANDARDS WE FOLLOW

TXCO manufactures and inspects flange requirements to the applicable international standard, customer drawing, purchase specification and project requirements.

Standard	Application
ASME B16.5	Pipe Flanges and Flanged Fittings
ASME B16.47	Large-Diameter Steel Flanges
ASME B16.48	Line Blanks
API 6A	Applicable wellhead and Christmas tree equipment requirements
MSS SP-44	Steel Pipeline Flanges
EN Standards	Applicable European requirements
DIN Standards	Applicable German requirements
JIS Standards	Applicable Japanese requirements
BS Standards	Applicable British requirements

The applicable standard and edition, pressure class, dimensions, tolerances, material specification, facing and inspection requirements are reviewed before manufacturing.

FLANGE MATERIALS

TXCO focuses on corrosion-resistant and high-performance flange materials. Carbon steel flange manufacturing is not part of TXCO's standard flange product range.

Material Group	Typical Grades / Materials
Austenitic Stainless Steel	F304, F304L, F316, F316L
High-Temperature Stainless Steel	F321, F347, F347H
Duplex Stainless Steel	UNS S31803, UNS S32205 / 2205
Super Duplex Stainless Steel	UNS S32750, UNS S32760 / 2507
Nickel Alloys	Inconel 600, 625, 718
Nickel-Iron-Chromium Alloys	Incoloy 800, 800H, 800HT, 825
Nickel-Copper Alloys	Monel grades
Special Alloys	Alloy 20 and other specified alloys

Material selection is based on operating pressure and temperature, corrosion environment, fluid characteristics, mechanical requirements, applicable material specification and customer or project requirements.

QUALITY ASSURANCE

Quality is controlled throughout the manufacturing process, from material receipt through machining, inspection and final release.

Material Control

Materials are verified against the specified grade and applicable material requirements. Heat-number traceability is maintained .

Dimensional Control

Critical dimensions are checked against the applicable standard or approved drawing, including outside diameter, bore, thickness, bolt-circle diameter, bolt-hole diameter, number of holes, hub dimensions and overall dimensions.

Machining Control

Critical machined features are controlled for accuracy, including flange faces, RTJ grooves, bores, bolt holes, bevels, critical diameters and surface condition.

Material Verification

Depending on project requirements, inspection will include PMI, chemical composition, hardness testing, mechanical testing and material certificate verification.

Final Inspection

Finished flanges are evaluated against the applicable standard, approved drawing, purchase order and inspection requirements before release.

TRACEABILITY & QUALITY DOCUMENTATION

Documentation is provided according to customer and project requirements and may include:

- Material Test Certificates
- EN 10204 3.1 Certificates
- EN 10204 3.2 where specified
- PMI Reports
- Dimensional Inspection Reports
- Hardness Reports
- NDE Reports
- Certificate of Conformance
- Inspection Reports
- Heat-number traceability
- Final inspection documentation

TXCO FLANGES & SEALING EXPERTISE

TXCO's flange capability is supported by our experience in industrial sealing products, including RTJ Gaskets, Spiral Wound Gaskets, Kammprofile Gaskets, Metal Jacketed Gaskets, Metallic Gaskets, Non-Metallic Gaskets, PTFE Gaskets and Flange Insulation Kits.

This allows TXCO to consider the complete bolted-joint system: FLANGE + GASKET + BOLTING + OPERATING CONDITIONS.

Our objective is to supply technically controlled flange solutions that meet defined requirements and support dependable sealing performance in demanding industrial service.

ENGINEERED & CUSTOM FLANGES

TXCO manufactures custom-made flanges to customer drawings and project-specific requirements where standard configurations do not meet the application.

Custom flange solutions can be developed for special dimensions, bore configurations, bolt patterns, thicknesses, facings, RTJ grooves, reducing arrangements, internal bevels and specified stainless, duplex, super duplex or nickel-based alloys.

Each custom requirement is reviewed against the approved drawing, applicable ASME/API or other governing standard, material specification and inspection requirements before manufacture.

TXCO

PRECISION ENGINEERED. STANDARDS DRIVEN. QUALITY CONTROLLED.

TXCO Sealing Products Private Limited combines precision manufacturing, material expertise, standards-based production and sealing knowledge to provide flange solutions for demanding industrial applications.

7. TXCO Valve Components

Precision Sealing Technology for Critical Flow Control & Zero-Emissions Integrity

A valve's operational integrity relies entirely on the reliability of its internal sealing interfaces. Under extreme pressure dynamics, severe thermal shock, and aggressive chemical exposure, substandard seals lead to stem leaks, seat blowouts, and non-compliant fugitive emissions. TXCO Valve Components are engineered to provide low-friction, high-integrity static and dynamic barriers within critical flow control equipment. Precision-manufactured to exacting tolerances, our components ensure bubble-tight shut-off, smooth actuator torque, and maximum operational lifespan across severe-service isolation and control valves.

Core Sealing Solutions

- **Precision O-Rings & Anti-Extrusion Backup Rings**

Molded from high-performance elastomers (FKM, HNBR, EPDM) and chemically inert Perfluoroelastomers (FFKM), TXCO O-rings provide primary static and dynamic sealing for valve stems, body joints, and internal cartridges. Certified for Rapid Gas Decompression (RGD / AED) under NORSOK M-710, they resist micro-blistering during rapid depressurization up to 10,000 PSI. Paired with PEEK or PTFE backup rings, they eliminate seal extrusion under severe differential pressures.

- **API 622 / ISO 15848-1 Low-Emission Gland Packing**

Functioning as the dynamic stem seal for rotating and reciprocating valves, TXCO gland packings are precision-braided from expanded flexible graphite, Inconel-wire reinforcement, carbon yarn, and modified ePTFE. Their self-lubricating, high-density construction maintains continuous radial contact against stuffing box walls, minimizing actuator breakout torque while exceeding strict fugitive emissions standards (API 622, TA-Luft, ISO 15848-1 Class A).

- **Valve Seats & Metallic / Soft Seat Inserts**

CNC-machined for bubble-tight shut-off in ball, butterfly, gate, and globe valves, TXCO seat rings deliver mechanical strength and thermal memory. Manufactured from engineering thermoplastics—including Virgin/Filled PTFE, PEEK, PCTFE, Devlon V, and Nylon—or solid metallic base rings, these seats resist creep deformation under heavy line loads, maintaining sealing integrity from cryogenic LNG conditions (-196°C) to elevated process temperatures (+260°C).

<u>Component</u>	<u>Primary Materials</u>	<u>Temperature Range</u>	<u>Pressure Rating</u>	<u>Key Industry Standards</u>
Precision O-Rings	FFKM, FKM, HNBR, EPDM	-50°C to +315°C	Up to 10,000 PSI	NORSOK M-710, ISO 23936-2
Gland Packing	ePTFE, Graphite, Inconel Wire	-200°C to +650°C	Class 150 – 2500	API 622, ISO 15848-1, TA-Luft
Seat Rings & Inserts	PEEK, PCTFE, Devlon V, PTFE	-196°C to +260°C	Class 150 – 2500	API 6D, ASME B16.34

8.TXCO Monolithic Insulation Joints

Definitive Dielectric Isolation & Structural Integrity for Buried and Aboveground Pipelines

Engineered as solid-state, fully welded structural forgings, **TXCO Monolithic Insulation Joints** deliver permanent Cathodic Protection (CP) isolation and zero-leakage fluid containment across critical pipeline networks. By integrating high-strength forged steel components with advanced dielectric insulation into a single boltless unit, TXCO monolithic joints eliminate field maintenance, withstand aggressive soil loading, and outlast the pipeline itself.

Engineering Highlights

- **Direct-Burial & Maintenance-Free:** Encased in factory-applied 100% solids polyurethane or liquid epoxy coatings, eliminating concrete inspection vaults and bolt retorquing.
- **Encapsulated Dual-Seal System:** Twin primary seals (FKM, HNBR, or spring-energized ePTFE) housed inside a void-free dielectric cavity protect sealing elements from fluid erosion and explosive decompression.
- **Full Bending Moment Resistance:** High-strength forged bodies (ASTM A694 / A350 LF2) match or exceed line-pipe yield strength, easily handling ground settlement, thermal growth, and soil overburden.
- **Guaranteed Dielectric Barrier:** High-density G10/G11 insulating rings ensure $\geq 100 \text{ M}\Omega$ electrical resistance and $\geq 5 \text{ kV}$ dielectric breakdown strength.

Engineering Specifications & Standards

Performance Parameter	Technical Specification
Applicable Design Standards	ASME B31.4, ASME B31.8, ASME Sec VIII Div 1/2, API 6D, NACE SP0169, ISO 15589-1
Nominal Sizes & Ratings	NPS 2" to 60" (DN50 – DN1500) ASME Class 150 to 2500 / API 2,000 to 10,000 PSI
Forging Metallurgy	ASTM A694 (F42 through F70), ASTM A105, ASTM A350 LF2, Duplex S31803 / Super Duplex S32750
Dielectric Performance	Electrical Resistance $\geq 100 \text{ M}\Omega$ at 1,000V DC; Dielectric Breakdown Strength $\geq 5 \text{ kV}$ (50 Hz)
Protective Coatings	Internal: Solvent-free epoxy lining External: Heavy-duty 100% solids polyurethane / epoxy coating
Quality Verification & NDE	100% Radiographic (RT), Ultrasonic (UT), Hydrostatic (1.5x Design Pressure), and High-Voltage Spark Testing

Strategic Field Deployments

- **Subterranean Mainline Insulation:** Directly buried into cross-country oil, gas, NGL, and slurry pipelines to establish precise Cathodic Protection (CP) zones without underground maintenance infrastructure.
- **Facility & Plant Boundaries:** Installed at refinery battery limits, compressor/pump stations, city gates, and tank farms to block stray electrical currents from entering plant grounding networks.
- **Offshore Risers & Shore Approaches:** Deployed at landfall tie-ins and subsea riser bases to isolate offshore platform galvanic systems from onshore pipeline networks.

About us

TXCO Sealing Products Private Limited – Engineering Absolute Reliability

Where extreme pressure, thermal shock, and volatile chemical media leave zero margin for error, precision is more than a specification—it is an absolute engineering duty. **TXCO Sealing Products Private Limited** manufactures mission-critical industrial gaskets, precision-forged flanges, monolithic insulation joints and severe-service valve components engineered to deliver complete end-to-end pipeline joint integrity across global oil & gas, petrochemical, nuclear, and power infrastructure. Operating at the intersection of advanced metallurgy, fluid containment science, and micro-inch metrology, our product line strictly adheres to ASME, API and all applicable parameters. We enforce strict hardness differentials to ensure plastic seal deformation while protecting expensive flange faces, guaranteeing fugitive-emission compliance (ISO 15848-1 / NACE MR0175) under extreme cyclic loads. Driven by a core mandate of **Zero Leaks, Zero Compromise, Zero Harm**—when system pressures reach their peak, TXCO delivers complete containment engineered from the inside out to protect human lives, safeguard critical assets, and preserve the environment.

Core Manufacturing & Pipeline Capabilities

- **Total Pipeline Solutions:** Complete joint containment and sealing assemblies engineered for end-to-end leak prevention and structural piping integrity.
- **Industrial Gaskets:** High-performance Ring Type Joint (RTJ R, RX, BX), Flange Isolation Kits, Spiral Wound, Kammprofile, graphite, and PTFE gaskets designed for severe thermal cycling.
- **Forged Flanges:** Weld Neck, Blind, Slip-On, and specialized flanges forged from, stainless steel, Duplex, and exotic alloys (Inconel, Hastelloy, Monel).
- **Valve Components:** Micro-machined internal trims, and seats built for low-torque, bubble-tight performance.

We measure our output not by the volume of steel turned, but by the trust placed in our pipeline solutions when system pressures reach their peak. Where total reliability is non-negotiable, TXCO delivers quality engineered from the inside out.

Vision

To be the premier global authority in complete pipeline sealing solutions—redefining benchmarks for engineering excellence, environmental stewardship, and zero-defect reliability across critical global infrastructure.

Mission

To engineer high-precision industrial gaskets, forged flanges, and valve components compliant with ASME, API 6A/6D, and ISO standards. We deliver custom, end-to-end joint integrity solutions that safeguard human lives, protect plant assets, and preserve the planet through uncompromised metallurgical science and a commitment to Zero Harm.

Core Values

- **Precision Engineering & Bespoke Design:** Meticulous attention to dimensions and tolerances, crafting tailor-made sealing solutions built for extreme operational demands.
- **Metallurgical Rigor & Quality:** Strict adherence to UNS specs, precise hardness limits (Brinell/Rockwell), 100% material traceability (EN 10204 3.1), and zero-void inspection.
- **Technical Mastery & Guidance:** Empowering partners through deep industry expertise, offering proactive technical advisory from material selection to installation.
- **Customer Partnership & Reliability:** Building lasting global relationships driven by exceptional service, on-time delivery, and dependable lifetime support.
- **On-Time Delivery** – We value our customers' time as much as our own. Every commitment carries responsibility, and we work to deliver on schedule, with the same quality and care that goes into every product.
- **Ethical Duty & Zero Harm:** Treating human safety and environmental protection not just as compliance goals, but as our core ethical responsibility.

Quality Policy

At TXCO Sealing Products Private Limited, quality is an absolute duty engineered into every industrial gasket, forged flange, and valve component to deliver zero-defect pipeline integrity in extreme-pressure environments. Operating under a "Right First Time, Every Time" philosophy, our manufacturing strictly aligns with ASME, API 6A/6D, NACE MR0175, and ISO 9001 standards. We enforce 100% Material Identification, strict Brinell/Rockwell hardness limits, EN 10204 3.1 heat-lot traceability, guarantee flaw-free metallurgical excellence. By pairing custom precision engineering with

expert technical support and dependable delivery, TXCO ensures every seal & Products which we manufacture protects human lives, safeguards critical infrastructure, and preserves the environment worldwide.

HSE policy

TXCO Sealing Products – Health, Safety, and Environmental (HSE) Policy Statement

At TXCO Sealing Products, high-performance engineering never comes at the cost of human safety or environmental integrity. We operate under one non-negotiable philosophy: **Zero Harm**—zero work-related injuries, zero occupational illnesses, and zero preventable environmental incidents.

Core HSE Pillars

- **Uncompromised Leadership & Accountability:** Executive leadership fully owns our HSE culture, embedding safety and sustainability directly into our operational strategy, resource allocation, and performance metrics.
- **Stop Work Authority:** Every employee and contractor has the absolute right, responsibility, and executive backing to halt any unsafe operation immediately, without fear of reprisal.
- **Proactive Hazard Elimination:** We utilize the Hierarchy of Controls to eliminate operational risks at the source, prioritizing engineering controls and safe process design over procedural workarounds.
- **Environmental Stewardship & Sustainability:** We minimize our ecological footprint through material efficiency across metals and polymers, manufacturing waste reduction, energy/water conservation, and responsible chemical management.
- **Global Standard Compliance:** We strictly adhere to all local and international HSE regulations, aligning our systems with global benchmarks including **ISO 45001** and **ISO 14001**.
- **Culture of Learning & Continuous Improvement:** We invest in ongoing HSE education, thorough incident investigations, transparent tracking, and routine third-party audits to raise industry standards every year.

Our Commitment to Stakeholders Integrity is manufactured into every seal we deliver. By maintaining a safe workplace and prioritizing environmental responsibility, we ensure our team returns home safely every day, our clients receive sustainable solutions, and our communities thrive alongside us.

services

TXCO Sealing Products Private Limited – Specialized Engineering & Field Services

We back our manufacturing with high-level metallurgical and mechanical engineering support. Designed around our core portfolio of industrial gaskets, forged flanges, and valve components, our engineering services optimize joint integrity, prevent fugitive emissions, and eliminate premature failure in critical piping systems.

1. Engineering Services & Flange Joint Integrity Calculations

- **ASME PCC-1 Compliant Torque & Tension Analysis:** We calculate targeted bolt loads, minimum gasket seating stress (y factor), and operational maintenance factors (m factor) to prevent yield under thermal cycling and pressure surges.
- **Fugitive Emission Control:** Sealing system optimization engineered to meet strict EPA and ISO 15848-1 leak rate thresholds.

2. Precision Reverse Engineering

- **Dimensional Synthesis & Profiling:** Utilizing precision manual metrology, profile gauges, and micrometers to map non-standard, obsolete, or worn gaskets and valve trim components (balls, seats, stems) without CAD drawings.
- **Metallurgical & Hardness Mapping:** On-site Positive Material Identification (PMI via portable XRF/OES spectro-analysis) and Brinell/Rockwell hardness testing. We ensure replacement RTJ rings maintain lower hardness than mating flange grooves (e.g., Soft Iron < 90 HB, SS 316 < 160 HB) to prevent expensive face galling.

3. Custom Gasket & Component Engineering

- **Bespoke Sealing Profiles:** Engineering non-standard gasket geometries, special serration pitches for Kammprofile cores, and composite layer arrangements (expanded graphite, virgin/modified PTFE, mica) for extreme media.
- **Metallurgical Selection for Severe Service:** Material specification matching UNS alloy designations (Inconel 625/825, Hastelloy C276, Super Duplex S31803, Monel 400) to operating media, compliant with **NACE MR0175 / ISO 15156** sour gas requirements.

4. Field Support & On-Site Flange Measurement

- **Flange Face & Serration Audits:** On-site verification of flange surface finishes (R_a / AARH ratings), phonographic serration depths, flange parallelism, and RTJ ring groove wear.
- **Leak Triage & Troubleshooting:** Rapid root-cause diagnosis of active joint leaks, seal blowout, or seat degradation to provide real-time field corrective actions during turnarounds and shutdowns.

5. Technical Training & Flange Management Workshops

- **ASME PCC-1 Bolting & Joint Assembly Procedures:** Practical maintenance training covering flange alignment tolerances, targeted bolt-stress selection, star-pattern torque sequencing, and nut-factor (μ) friction variables to ensure uniform gasket seating.

- **Flange-to-Pipe Fitment & Structural Alignment:** Specialized instruction on pipe-to-flange fitment, joint gap spacing, angular/axial offset correction, and stress-free bolt hole orientation during piping fabrication.
- **Root-Cause Failure Analysis (RCFA):** Technical diagnostic seminars for engineering teams to evaluate joint failure mechanisms, including inner-ring buckling, gasket crushing, chemical degradation, thermal stress relaxation, and flange face wire-drawing.

Industries : **chemical industries**

1. Pharmaceutical & FDA

Sanitary processing requires non-contaminating, zero-extractable sealing systems that endure aggressive Clean-In-Place (CIP) and Steam-In-Place (SIP) sterilization cycles. Gaskets and flanges must maintain structural integrity under thermal shock without leeching compounds into pure media.

Compliance with **FDA 21 CFR 177.1550, USP Class VI, ASME BPE, and 3-A Sanitary Standards** is mandatory.

- **Primary Application Areas:** Water for Injection (WFI) loops, API synthesis reactors, fermentation vessels, sterile fluid transfer lines, bio-reactors, and high-pressure utility steam lines.
- **Common Flange Types & Standards:**
 - **Weld Neck Flanges (ASME B16.5 WN Class 150/300):** Deployed on high-pressure clean steam boilers, WFI thermal loops, and plant utility headers (with internally polished bores).
 - **Slip-On Flanges (ASME B16.5 SO):** Restricted to non-critical utility lines, chilled water loops, and plant HVAC infrastructure.
- **Engineered Gasket & Sealing Selection:**
 - **Sanitary Tri-Clamp Gaskets:** Precision-molded in Virgin PTFE, USP Class VI EPDM, FKM, or Platinum-Cured Silicone for sanitary quick-disconnect unions.
 - **Virgin & Expanded PTFE (ePTFE) Gaskets:** Inert, zero-extractable sheet seals for high-purity process and chemical feed lines.
 - **PTFE Envelope Gaskets:** Elastomeric core encased in a seamless virgin PTFE shield for combined compressibility and chemical purity.
 - **Stainless Steel RTJ Gaskets (ASME B16.20 R Octagonal):** Precision 316L SS RTJs deployed in high-pressure plant utility boilers and extreme-pressure clean-steam generation headers.
 - **Flange Isolation Kits (FIK – TX-ISOPRO):** Full-face phenolic or G10 GRE isolation kits installed on utility boundary lines and raw water intake headers to block stray electrical currents.

2. Plastics & Polymers

Polymerization infrastructure operates under severe mechanical pressures, extreme thermal loading, and continuous exposure to molten resins, synthetic monomers, and heat transfer fluids (HTFs/Dowtherm). Sealing solutions must endure operating temperatures exceeding 500°C–900°C and pressures over 1,000 bar without thermal degradation, graphite oxidation, or joint relaxation under intense thermal expansion cycles. **Primary Application Areas:** High-Pressure Polyethylene (LDPE) hyper-compressors, polymerization reactors, monomer recovery units, extruder die heads, polymer melt pumps, and synthetic resin transfer lines.

- **Common Flange Types & Standards:**
 - **High-Pressure Weld Neck Flanges (ASME B16.5 WN Class 600 to 2500):** Essential for hyper-compressor discharge lines, polymer melt piping, and high-pressure polymerization loops.

- **RTJ Weld Neck Flanges:** Machined with API/ASME ring grooves for high-temperature, high-pressure polymer melt piping.
- **Engineered Gasket & Sealing Selection:**
 - **API 6A / ASME B16.20 Ring Type Joint (RTJ) Gaskets:** Heavy-duty machined **RX and BX profiles** in Soft Iron (Code D), Low Carbon Steel (Code S), or SS 316/304 engineered to contain extreme hydraulic pressures in hyper-compressors and melt pumps.
 - **High-Temperature Mica-Graphite Composite Gaskets:** Layered mica-graphite construction engineered for severe thermal oxidation environments up to 1000°C.
 - **Corrugated / Kammprofile Metal Gaskets with Graphite Facings:** High-recovery seals built for heat-transfer fluid loops and reactor shell joints.
 - **High-Temperature Flange Isolation Kits (FIK – TX-ISOPRO):** High-temp G11 GRE retainers with spring-energized PTFE or mica seals installed on electric heater flanges and HTF loops to prevent galvanic reaction.

3. Petrochemical & Refining

Refinery infrastructure deals with high-pressure, high-temperature (HPHT) hydrocarbons, volatile hydrogen gas, and sour gas (H_2S) service governed by **NACE MR0175 / ISO 15156**. Sealing systems must provide fire-safe blowout protection (**API 6FB / ISO 10497**) and comply with stringent low-emission mandates (**ISO 15848-1 / TA-Luft**) under severe pipeline vibration and pressure surges up to 20,000 PSI.

- **Primary Application Areas:** API 6A high-pressure wellhead Christmas trees, Hydrocracker units, Crude Distillation Units (CDU/VDU), Fluid Catalytic Cracking Units (FCCU), high-pressure flare manifolds, and submarine risers.
- **Common Flange Types & Standards:**
 - **Weld Neck Flanges (ASME B16.5 / B16.47 WN - RF & RTJ Facings):** Standard across all high-pressure, severe-service refinery process piping.
 - **Long Weld Neck Flanges (LWN):** Preferred for vessel nozzles, reactor headers, and column connections to eliminate girth weld stress concentration.
 - **API 6A Flanges (Type 6B & 6BX):** Heavy-duty 2,000 to 20,000 PSI rated flanges used on wellheads, blowout preventers (BOPs), and high-pressure manifolds.
 - **Orifice Flanges (ASME B16.36):** Specialized WN flanged unions equipped with differential pressure tapping holes for refinery flow metering.
 - **Blind Flanges (ASME B16.5 BL):** Used for line blinding during turnarounds, hydrotesting, and manifold capping.
- **Engineered Gasket & Sealing Selection:**

- **API Spec 6A & ASME B16.20 Ring Type Joint (RTJ) Gaskets:** Flagship solid metal R (**Oval & Octagonal**), RX, and BX rings manufactured in 316L, Super Duplex, Inconel 625, or Alloy 825 with strict Brinell hardness controls.
- **ASME B16.20 Spiral Wound Gaskets (SWG):** Built with internal/external guide rings in 316L or exotic alloys with high-density flexible graphite filler to prevent inner-ring buckling.
- **Kammprofile Gaskets with Inner Rings:** Heavy-duty serrated metal cores engineered for refinery heat-exchanger main shell-to-tube joints.
- **API 6FB Fire-Safe Flange Isolation Kits (FIK – TX-ISOPRO):** Critical Cathodic Protection (CP) isolation kits featuring G10/G11 GRE retainers, spring-energized PTFE seals, and full-length insulating sleeves for pipelines, tank batteries, and subsea risers.

4. Chemical Processing

General and specialty chemical plants utilize continuous and batch-processing systems cycling between full vacuum, high pressures, and wide temperature swings. Gaskets must withstand broad chemical exposure, prevent joint relaxation on non-rigid or glass-lined flanges, and maintain low bolt-load sealing performance while providing blowout protection on high-pressure headers.

- **Primary Application Areas:** High-pressure reaction loops, gasification units, distillation columns, acid feed manifolds, scrubbers, evaporators, flanged pump housings, and plant-wide high-pressure utility headers.
- **Common Flange Types & Standards:**
 - **Weld Neck Flanges (ASME B16.5 WN):** Used for main chemical process headers, high-pressure utility steam, and hazardous media transfer.
 - **Slip-On Flanges (ASME B16.5 SO):** Deployed on low-pressure, non-hazardous process lines and utility water loops.
 - **Socket Weld Flanges (ASME B16.5 SW):** Utilized on small-bore high-pressure chemical dosing and instrument piping lines.
 - **Threaded Flanges (ASME B16.5 TH):** Restricted to low-pressure utility lines where welding is prohibited.
- **Engineered Gasket & Sealing Selection:**
 - **Ring Type Joint (RTJ) Gaskets:** Deployed in high-pressure reaction loops, high-temperature gasifiers, and extreme-pressure chemical transport manifolds.
 - **Tanged Metal-Reinforced Graphite Gaskets:** High-strength flexible graphite sheets bonded to a 316L tanged steel core for blowout protection on standard RF flanges.
 - **Low-Emission (Low-E) Spiral Wound Gaskets:** Formulated with high-density filler matrices to meet ISO 15848-1 Class A leak rates.

- **Modified PTFE Sheet Gaskets:** Biaxially oriented PTFE filled with silica, barium sulfate, or hollow glass microspheres for low bolt-load and glass-lined steel applications.
- **Flange Isolation Kits (FIK – TX-ISOPRO):** G10 GRE retainers with dual elastomeric (FKM/NBR) or ePTFE seal elements, installed at plant battery limits, tank connections, and underground entries.

Energy & Oil & Gas

1. Upstream & Offshore

Operating in harsh marine environments, subsea depths, and High-Pressure High-Temperature (HPHT) wellhead zones demands ultimate mechanical reliability. Equipment is subjected to hydrostatic ocean pressures, intense structural flexure, dynamic topside vibration, sour gas (H_2S) exposure under **NACE MR0175 / ISO 15156**, and wellhead working pressures reaching up to 20,000 PSI.

- **Who We Serve:** Exploration & Production (E&P) operators, offshore drilling contractors, FPSO operators, subsea tree and Blowout Preventer (BOP) OEMs, subsea piping EPCs, and naval marine infrastructure contractors.
- **Primary Application Areas:** API 6A wellhead Christmas trees, subsea BOP stack connections, offshore production risers, subsea manifolds, choke-and-kill manifolds, high-pressure mud pumps, and topside separation units.
- **Common Flange Types & Standards:**
 - **API 6A Type 6B & 6BX Flanges:** Heavy-duty, extreme-pressure flanges engineered for wellhead, BOP, and Christmas tree assemblies.
 - **ASME B16.5 / B16.47 Weld Neck Flanges (WN - RTJ Facing):** Standard high-pressure connections for topside process manifolds and riser piping.
- **Engineered Gasket & Sealing Selection:**
 - **API Spec 6A & 17D Ring Type Joint (RTJ) Gaskets:** Solid metal **RX, BX, and pressure-balanced SRX / SBX subsea profiles** machined in Inconel 718, Super Duplex S32750, and 316L SS with strict hardness verification to force plastic seal deformation without damaging expensive groove faces.
 - **API 6FB Fire-Safe Flange Isolation Kits (FIK -TX-ISOPRO):** High-dielectric G10/G11 Glass-Reinforced Epoxy (GRE) retainers with spring-energized PTFE seal elements engineered for splash-zone atmospheric protection and subsea Cathodic Protection (CP) isolation.
 - **Compact Flange IX Metallic Seals:** Machined metallic seal rings engineered for NORSOK L-005 compact flanges to provide dual-cone seating under severe dynamic bending moments.

- **Severe-Service Spiral Wound Gaskets (ASME B16.20):** Winding metallurgy in Inconel 625 or 316L with high-density flexible graphite or ePTFE fillers for topside gas compression and separation units.

2. Midstream & Pipeline

Transporting crude oil, natural gas, NGLs, and refined products across thousands of miles of cross-country and subsea pipelines requires long-term joint integrity. Pipeline infrastructure must withstand continuous ground movement, severe thermal expansion, soil load stresses, pigging impact forces, and electrochemical corrosion—all while complying with strict environmental leakage regulations.

- **Who We Serve:** Interstate crude and gas pipeline operators, compressor and pump station EPC contractors, metering skid manufacturers, gas processing plant operators, underground storage facility managers, and pipeline maintenance providers.
- **Primary Application Areas:** Cross-country transmission pipelines, gas compressor stations, liquid pump stations, pig launchers and receivers, metering/custody transfer skids, underground valve vaults, and tank farm battery limits.
- **Common Flange Types & Standards:**
 - **High-Yield Weld Neck Flanges (MSS SP-44 / CSA Z245.12 / ASME B16.47):** High-strength forged flanges matched to high-yield line pipe grades (API 5L X52 through X80).
 - **ASME B16.36 Orifice Flanges:** Flanged unions equipped with integrated pressure tapings for custody transfer flow measurement.
 - **Blind Flanges (ASME B16.5 BL):** Deployed on pig trap quick-actuating closures and maintenance bypass lines.
- **Engineered Gasket & Sealing Selection:**
 - **Cathodic Protection Flange Isolation Kits (FIK -TX-ISOPRO):**
 - **Low-Emission Spiral Wound Gaskets (ASME B16.20 Low-E):** Specially engineered filler density SWGs designed to maintain tightness under remote compressor station vibration, achieving **ISO 15848-1 Class A** fugitive emission compliance.
 - **Kammprofile Gaskets with Inner & Outer Rings:** Heavy-duty serrated metal cores faced with expanded flexible graphite for high-vibration pump station discharge flanges.
 - **API Spec 6A Ring Type Joint (RTJ) Gaskets:** Solid metal R Octagonal and RX rings used on high-pressure gas gathering manifolds and pig trap doors.

3. Downstream Refining

Refining operations represent one of the most demanding thermal and chemical environments in heavy industry. Fluid containment systems must endure high thermal shock, continuous operating temperatures exceeding 650°C, High-Temperature Hydrogen Attack (HTHA), toxic Hydrofluoric (HF)

acid alkylation, and sour hydrocarbon media while adhering to strict **EPA Method 21** and **TA-Luft / ISO 15848-1** zero-leakage mandates.

- **Who We Serve:** Integrated petroleum refineries, petrochemical complexes, independent fuel processors, refinery turnaround and maintenance contractors, heat exchanger OEMs, and industrial overhaul service teams.
- **Primary Application Areas:** Crude Distillation Units (CDU/VDU), Fluid Catalytic Cracking Units (FCCU), Hydrocrackers, Hydrotreaters, HF/Sulfuric Acid Alkylation units, Delayed Cokers, Flare Headers, and Shell-and-Tube Heat Exchangers.
- **Common Flange Types & Standards:**
 - **Weld Neck Flanges (ASME B16.5 / B16.47 WN - RF & RTJ):** Standard across all high-pressure, high-temperature refinery units to ensure smooth stress transition from flange to pipe.
 - **Long Weld Neck Flanges (LWN):** Preferred for heavy vessel nozzles, column manways, and reactor nozzles to eliminate girth-weld fatigue.
 - **Orifice Flanges (ASME B16.36):** Used for precise flow monitoring across refinery processing units.
- **Engineered Gasket & Sealing Selection:**
 - **ASME B16.20 Spiral Wound Gaskets (SWG):** Constructed with 316L, Inconel 625, or Monel 400 windings and high-density flexible graphite fillers. Features mandatory solid inner rings to prevent radial inner-ring buckling under severe thermal shock.
 - **Kammprofile Gaskets for Heat Exchangers:** Serrated metallic cores faced with expanded flexible graphite or mica, custom-engineered for floating heads, channel covers, and shell-to-tubesheet joints subject to differential thermal expansion.
 - **API Spec 6A & ASME B16.20 Ring Type Joint (RTJ) Gaskets:** Precision-machined R, RX, and BX rings with strict Brinell hardness caps (e.g., Soft Iron < 90 HB, SS 316 < 160 HB) to protect expensive flange grooves from galling during cyclic operation.
 - **Monel-Wound & ePTFE Gaskets for HF Alkylation:** Specially designed seals using Monel 400 cores/windings and virgin ePTFE to resist lethal hydrofluoric acid corrosion and prevent crevice attack.
 - **API 6FB Fire-Safe Flange Isolation Kits(FIK – TX-CERAMPRO)** Installed at tank battery limits, offsite transfer lines, and dissimilar metal vessel connections for CP isolation and fire-safe joint integrity.

Manufacturing & Heavy Industry

1. Food & Beverage (FDA)

Food and beverage processing requires sanitary, non-contaminating, and zero-extractable fluid containment systems that withstand high-temperature processing, organic fats, acidic flavorings, and aggressive Clean-In-Place (CIP) and Steam-In-Place (SIP) chemical sterilization cycles. Sealing materials must endure continuous thermal shock without leeching plasticizers, degrading, or creating internal crevices that encourage bacterial growth. Strict compliance with **FDA 21 CFR 177.1550 / 177.2600, USP Class VI, ASME BPE, and 3-A Sanitary Standards** is mandatory.

- **Who We Serve:** Industrial food processors, commercial beverage & bottling complexes, dairy & milk processing plants, breweries, distilleries, sanitary equipment OEMs (mixers, homogenizers, pasteurizers), and bioprocess piping contractors.
- **Primary Application Areas:** High-temperature short-time (HTST) pasteurization loops, aseptic filling lines, CIP/SIP utility distribution headers, mixing kettles, high-pressure homogenizers, evaporation vessels, and steam sterilization headers.
- **Common Flange Types & Standards:**
 - **Weld Neck Flanges (ASME B16.5 WN Class 150/300):** Deployed on high-pressure utility steam lines, thermal fluid loops, and boiler feed systems (with internally ground and polished welds).
 - **Slip-On & Threaded Flanges (ASME B16.5 SO/TH):** Restricted strictly to non-process utility lines, raw water intake, and facility HVAC cooling networks.
- **Engineered Gasket & Sealing Selection:**
 - **Restructured Virgin & Expanded PTFE (ePTFE) Sheets:** Inert, non-contaminating sheet seals designed for flanged product transfer lines, offering zero creep relaxation under thermal cycling.
 - **PTFE Envelope Gaskets:** Combining an elastomeric core (EPDM/FKM) with a seamless virgin PTFE outer sheath for enhanced compressibility on low-bolt-load sanitary vessel flanges.
 - **Stainless Steel RTJ Gaskets (ASME B16.20 R Octagonal):** Precision 316L SS Ring Type Joints deployed in high-pressure clean-steam generation boilers and thermal sterilization utility headers.
 - **Flange Isolation Kits (FIK – TX-ISOPRO):** Full-face phenolic or G10 GRE isolation kits installed on facility boundary water mains and utility supply piping to isolate stray electrical currents and protect plant grounding systems.

2. Heavy Machinery & Steel

Heavy machinery and steel manufacturing represent an environment of extreme mechanical stress, intense radiant heat, shock loading, and massive hydraulic pressures. Equipment must endure continuous exposure to molten metal proximity, abrasive furnace scale, aggressive cooling water, and hydraulic working pressures often exceeding 350 bar (5,000 PSI). Sealing integrity depends on robust

metallurgy, high-recovery gasket geometries, and exceptional resistance to thermal oxidation up to 1,000°C.

- **Who We Serve:** Integrated steel mills (Blast Furnace, BOF, EAF), hot/cold rolling mills, heavy hydraulic press OEMs, metal forging facilities, smelting operations, mining equipment builders, and heavy industrial maintenance teams.
- **Primary Application Areas:** Blast furnace cooling plates, oxygen steelmaking lances, continuous casting machines, hot rolling mill hydraulic power units (HPUs), heavy forging presses, slag granulators, and high-pressure steam descaling headers.
- **Common Flange Types & Standards:**
 - **High-Pressure Weld Neck Flanges (ASME B16.5 / B16.47 WN Class 300 to 2500):** Essential for high-pressure cooling water mains, oxygen supply lines, and descaling spray headers.
 - **Custom Large-Diameter Vessel Flanges:** Non-standard square, rectangular, or circular flanged joints built into furnace doors, gear drives, and heat treatment shells.
 - **Threaded & Socket Weld Flanges (ASME B16.5 SW/TH):** Utilized on small-bore high-pressure grease lines, hydraulic control lines, and sensor wells.
- **Engineered Gasket & Sealing Selection:**
 - **High-Temperature Flange Isolation Kits (FIK – TX-CERAMPRO)**
 - **API Spec 6A & ASME B16.20 Ring Type Joint (RTJ) Gaskets:** Heavy-duty solid metal **R (Oval/Octagonal) and RX profiles** in Soft Iron (Code D) or SS 304/316 engineered for extreme-pressure hydraulic systems, oxygen lances, and forge press manifolds.
 - **Heavy-Duty Spiral Wound & Kammprofile Gaskets:** Metal-cored seals with tanged graphite facings engineered to survive heavy structural vibration on rolling mill cooling water piping and high-pressure steam headers.
 - **Heavy-Duty Flange Isolation Kits (FIK – TX ISOPRO):** High-strength G10/G11 GRE isolation sets installed on electric furnace transformers, electrolysis cells, and underground cooling water mains to block massive stray electrical currents and prevent rapid galvanic destruction.

Metals & Mining

High-Pressure Acid Leach (HPAL) & Slurry Pipelines

Hydrometallurgical mineral extraction and long-distance ore transport represent two of the most aggressive mechanical and chemical processing environments in heavy industry. Fluid containment systems must concurrently survive extreme erosion-corrosion from high-velocity mineral slurries (silica, nickel laterite, copper tailings), concentrated hot sulfuric acid (H_2SO_4), continuous hydraulic shock, thermal expansion up to 270°C, and system pressures exceeding 200 bar (3,000 PSI).

- **Who We Serve:** Nickel, cobalt, gold, and lithium hydrometallurgical processing complexes, long-distance slurry transmission pipeline operators, mineral tailings management facilities (TMF), autoclave vessel OEMs, slurry pump manufacturers, and global mining EPC contractors.
- **Primary Application Areas:** HPAL titanium autoclave reactors, high-pressure slurry feed pumps, acid injection lines, flash vessel discharge headers, choke manifold assemblies, long-distance iron ore/copper slurry transmission pipelines, and tailings disposal lines.
- **Common Flange Types & Standards:**
 - **Weld Neck Flanges (ASME B16.5 / B16.47 WN Class 600 to 2500):** Primary choice for severe-service, high-pressure slurry discharge lines to ensure smooth bore alignment and eliminate turbulence-induced wall erosion.
 - **Titanium & Exotic Alloy Lap Joint Flanges (ASME B16.5 LJ):** Titanium Grade 2, Grade 7, or Hastelloy C276 stub ends paired with carbon steel backup flanges to safely handle hot acid lines at optimized material costs.
 - **Custom Lined Pipe Flanges:** Specialized flanged joints engineered to accommodate internal rubber, polyurethane, or basalt erosion-resistant linings without pinching or shearing the internal barrier.
- **Engineered Gasket & Sealing Selection:**
 - **Exotic Alloy Ring Type Joint (RTJ) Gaskets (ASME B16.20):** Precision-machined **R, RX, and BX rings** in Titanium Grade 2/7, Hastelloy C276, or Super Duplex S32750 engineered for HPAL titanium autoclaves, acid injection nozzles, and high-pressure slurry isolation valves.
 - **Erosion-Resistant Kammprofile Gaskets:** Titanium or Hastelloy serrated metallic cores faced with expanded ePTFE or flexible graphite, engineered to resist internal fluid scouring and maintain joint tightness across high-vibration slurry pump manifolds.
 - **Restructured ePTFE Sheet & Envelope Seals:** Biaxially oriented ePTFE sheets resistant to full pH 0–14 acid exposure, formulated to eliminate cold-flow under cyclic thermal loading on low-to-medium pressure acid feed piping.
 - **Heavy-Duty Cathodic Protection Flange Isolation Kits (FIK -TX ISOPRO):** High-dielectric G10/G11 Glass-Reinforced Epoxy with TX-FLEX ,seals to block galvanic corrosion cells between titanium/exotic vessel interfaces and steel piping, while maintaining Cathodic Protection (CP) integrity on cross-country slurry pipelines.